STAT 3004

Mr.
Xavier Zhang
The University of Queensland

25 Semester 1

Contents

1 Branching Process **2**
1.1 Probability Generating Function . . . 2
1.2 Branching processes . . . 3

2 Random Walks **9**
2.1 Introduction to Random Walk . . . 9
2.2 Simple Symmetric Random Walk . . . 11
2.3 Simple Random Walk . . . 12

3 Discrete Time Markov Chains **15**
3.1 Stochastic process . . . 15
3.2 Introduction to DTMC . . . 15
3.3 Limiting Distribution . . . 17
3.3.1 Time–homogeneous DTMC . . . 17
3.3.2 Finding Limiting Distribution . . . 17
3.3.3 Represent the Transition Matrix . . . 20
3.3.4 Example . . . 20
3.4 Communication classes . . . 21
3.5 Global and local balance . . . 24
3.6 Reserve chain . . . 28

4 Poisson Processes **29**
4.1 Counting processes . . . 29
4.2 Time-homogeneous Poisson process . . . 29
4.3 Non-homogeneous Poisson process . . . 33
4.4 Spatial Poisson Process (Poisson Random Measure) . . . 36

5 (Continuous Time) Markov Chains **38**
5.1 Definition and Fundamental Results . . . 38
5.2 Q Matrix . . . 41
5.3 Limiting Distributions . . . 42
5.4 Classification of States for CTMCs . . . 44
5.5 Birth-Death Processes . . . 47

6 Probabilistic Measure Theory **50**
6.1 Preliminary . . . 50
6.2 The Probability Space . . . 51
6.2.1 The σ-algebra . . . 51
6.2.2 The Probability Measure . . . 54
6.2.3 Borel-Cantelli Lemmas . . . 55
6.3 Random Variables . . . 57
6.3.1 The Random Variable and its Distribution . . . 57
6.3.2 Types of Random Variables . . . 59
6.3.3 Expectation . . . 61
6.3.4 Convergence . . . 63
6.3.5 Law of Large Numbers . . . 66

Chapter 1

Branching Process

1.1 Probability Generating Function

Definition 1.1.1

Let X be a discrete random variable taking values in $\{0, 1, 2, \dots\}$. The *probability generating function* (pgf) of X is the function $G : [0, 1] \to [0, 1]$ defined by

$$G(z) \;:=\; \mathbb{E}z^X \;=\; \sum_{k=0}^{\infty} z^k \, \mathbb{P}(X = k).$$

G is continuous on $[0, 1]$ and continuously differentiable on $(0, 1)$.

Note:

$$G(1) = \mathbb{E}[1^X] = 1.$$

$$G(z) = \mathbb{E}[z^X] = \sum_{k=0}^{\infty} z^k P(X = k)$$

$$G'(z) = \sum_{k=0}^{\infty} k z^{k-1} P(X = k)$$

Fact 1.1.2

- $G(0) = P(X = 0)$ (more generally, $P(X = k) = \frac{G^{(k)}(0)}{k!}$)
- $G'(1) = \mathbb{E}[X]$
- $G''(1) = \mathbb{E}[X(X - 1)]$
- $\mathrm{Var}(X) = G''(1) + G'(1) - (G'(1))^2$
- $G(\bar{z})$ exists for $0 \le \bar{z} < 1$

Example.

> For $X \sim \mathrm{Poi}(\lambda)$, $\lambda > 0$ and $P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$, $\quad k = 0, 1, 2, \dots$
> Find $G(z) = \mathbb{E}\left[z^X\right]$

$$G(z) = \sum_{k=0}^{\infty} z^k \cdot e^{-\lambda} \frac{\lambda^k}{k!}$$

$$= e^{-\lambda} \sum_{k=0}^{\infty} \frac{(z\lambda)^k}{k!}$$

Using the power series expansion of the exponential function:

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

we get

$$\begin{aligned} G(z) &= e^{-\lambda} e^{z\lambda} \\ &= e^{-\lambda(1-z)} \end{aligned}$$

Example.

> For $X \sim \text{Geom}(p)$, $\quad p \in (0,1]$ and $P(X=k) = (1-p)^{k-1}p, \quad k = 1, 2, \ldots$
> Find $G(z) = \mathbb{E}\left[z^X\right]$

$$G_X(z) = \sum_{k=1}^{\infty} z^k (1-p)^{k-1} p$$

Rewriting the summation:

$$\begin{aligned} &= \sum_{k=0}^{\infty} z^{k+1} (1-p)^k p \\ &= zp \sum_{k=0}^{\infty} (z(1-p))^k \end{aligned}$$

Using the formula for the sum of an infinite geometric series:

$$\sum_{k=0}^{\infty} r^k = \frac{1}{1-r}, \quad |r| < 1$$

we obtain:

$$G_X(z) = \frac{zp}{1 - z(1-p)}.$$

Alternatively, defining $X_0 = X - 1$ such that $X_0 \sim \text{Geom}_0(p)$, we find:

$$\begin{aligned} G_{X_0}(z) &= \mathbb{E}[z^{X_0}] = \mathbb{E}[z^{X-1}] \\ &= z^{-1}\mathbb{E}[z^X] = z^{-1} G_X(z). \end{aligned}$$

Substituting $G_X(z)$:

$$G_{X_0}(z) = \frac{p}{1 - z(1-p)}.$$

Fact 1.1.3

- $\sum_{k=0}^{m} r^k = \frac{1-r^{m+1}}{1-r}, \quad r \neq 1$
- $\sum_{k=0}^{\infty} r^k = \frac{1}{1-r}, \quad |r| < 1$

1.2 Branching processes

Definition 1.2.1: Branching process

A **branching process** models the evolution of a *stochastic system* where individuals produce a **random number** of offspring in each generation.

Let S_n denote the number of individuals in the nth generation, where $n = 0, 1, 2, \ldots$, and set $S_0 = 1$. Define $X_{i,n}$ as the number of offspring of the ith individual in the nth generation.

Example.

$$X_{1,0} = 2$$
$$X_{1,1} = 2, \quad X_{2,1} = 1$$
$$X_{1,2} = 3, \quad X_{2,2} = 0, \quad X_{3,2} = 0$$

The total number of individuals in each generation is:

$$S_0 = 1, \quad S_1 = 2, \quad S_2 = 3, \quad S_3 = 3.$$

All $X_{i,n}$ are **i.i.d.** random variables. Denote by X a generic copy of $X_{i,n}$, and let $G(z)$ be its probability generating function (pgf). Write $\mu = \mathbb{E}[X], \quad \sigma^2 = \mathrm{Var}(X)$ (if they exist).

Fact 1.2.2

In general, $\quad S_n = \sum_{i=1}^{S_{n-1}} X_{i,n-1}$.

Q: Will the population become *extinct* or does it have a chance of *persisting*?

$$\{\text{extinct}\} \iff \exists m \geq 1, \text{ s.t. } S_m = 0$$

Fact 1.2.3

- $\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X|Y]]$
- If X and Y are independent $\to \mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$
- $\mathbb{E}[g(X)h(Y)|X] = g(X)\mathbb{E}[h(Y)|X] = g(X)\mathbb{E}[h(Y)]$ if X, Y are independent.

Sub Q: $\mathbb{E}[S_n] = ?$

$$\mathbb{E}[S_n] = \mathbb{E}\left[\sum_{i=1}^{S_{n-1}} X_{i,n-1}\right] = \mathbb{E}\left[\mathbb{E}\left[\sum_{i=1}^{S_{n-1}} X_{i,n-1}\middle| S_{n-1}\right]\right]$$

$$= \mathbb{E}\left[\sum_{i=1}^{S_{n-1}} \mathbb{E}[X_{i,n-1}|S_{n-1}]\right] = \mathbb{E}\left[\sum_{i=1}^{S_{n-1}} \mathbb{E}[X_{i,n-1}]\right] \text{ since } X_{i,n-1} \text{ are independent.}$$

$$= \mathbb{E}\left[\sum_{i=1}^{S_{n-1}} \mu\right] = \mathbb{E}\left[S_{n-1}\mu\right] = \mu\mathbb{E}[S_{n-1}]$$

Since $S_0 = 1$ (but not always!),

$$\mathbb{E}[S_1] = S_0\mu, \quad \mathbb{E}[S_2] = S_0\mu^2, \quad \ldots, \quad \mathbb{E}[S_n] = S_0\mu^n.$$

Note:

- $\mu \geq 0$.
- $\mu < 1 \quad \Rightarrow \quad \mathbb{E}[S_n] \to 0 \quad \text{as } n \to \infty$
- $\mu = 1 \quad \Rightarrow \quad \mathbb{E}[S_n] = S_0 = 1 \quad \forall n$
- $\mu > 1 \quad \Rightarrow \quad \mathbb{E}[S_n] \to \infty \quad \text{as } n \to \infty$

Fact 1.2.4

$$\mathrm{Var}(X) = \mathbb{E}[\mathrm{Var}(X \mid Y)] + \mathrm{Var}(\mathbb{E}[X \mid Y]).$$

Sub Q: $\mathrm{Var}(S_n) =?$

$$\mathrm{Var}(S_n) = \mathbb{E}[\mathrm{Var}(S_n|S_{n-1})] + \mathrm{Var}(\mathbb{E}[S_n|S_{n-1}])$$

Step (i)

$$\mathrm{Var}(S_n|S_{n-1}) = \mathrm{Var}\left(\sum_{i=1}^{S_{n-1}} X_{i,n-1}\Big|S_{n-1}\right)$$

Since the random variables $X_{i,n-1}$ are independent and identically distributed (i.i.d.) with variance σ^2, we use the variance property:

$$\mathrm{Var}\left(\sum_{i=1}^{m} X_i\right) = m \cdot \mathrm{Var}(X_i), \quad \text{for independent } X_i$$

Applying this to our case where $m = S_{n-1}$:

$$\mathrm{Var}(S_n|S_{n-1}) = S_{n-1} \cdot \sigma^2.$$

Taking expectation:

$$\mathbb{E}\left[S_{n-1}\sigma^2\right] = \sigma^2\mathbb{E}[S_{n-1}] = \sigma^2\mu^{n-1}, \quad n = 1, 2, \ldots$$

Step (ii)

$$\mathbb{E}[S_n|S_{n-1}] = \mu S_{n-1}$$

$$\mathrm{Var}(\mu S_{n-1}) = \mu^2\mathrm{Var}(S_{n-1})$$

Combining Results:

$$\mathrm{Var}(S_n) = \sigma^2\mu^{n-1} + \mu^2\mathrm{Var}(S_{n-1})$$

Expanding iteratively:

$$= \sigma^2\mu^{n-1} + \mu^2\left(\mu^{n-2}\sigma^2 + \mu^2\mathrm{Var}(S_{n-2})\right)$$

$$= \sigma^2\mu^{n-1}(1 + \mu + \mu^2 + \cdots + \mu^{n-1})$$

Using geometric series sum formula:

$$\mathrm{Var}(S_n) = \begin{cases} \sigma^2\mu^{n-1}\frac{1-\mu^n}{1-\mu}, & \mu \neq 1 \\ \sigma^2 n, & \mu = 1 \end{cases}$$

Note:

- $\mu < 1, \quad \mathrm{Var}(S_n) \to 0 \quad \text{as } n \to \infty$
- $\mu \geq 1, \quad \mathrm{Var}(S_n) \to \infty \quad \text{as } n \to \infty$

Definition 1.2.5: Probability of extinction

Let η_n be the probability of extinction at or before the nth generation:

$$\eta_n = P(\text{Extinct at or before } n^{\text{th}} \text{ Gen}) = P(S_n = 0)$$

Define η as the probability of *ultimate extinction*:

$$\eta = \lim_{n\to\infty} \eta_n.$$

Lemma 1.2.6: Markov's Inequality

For any random variable $X \geq 0$, and any $a > 0$,

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

Proof.

$$\begin{aligned}
E[X] &= E[X\mathbb{1}(X < a) + X\mathbb{1}(X \geq a)] \\
&= E[X\mathbb{1}(X < a)] + E[X\mathbb{1}(X \geq a)] \\
&\geq E[X\mathbb{1}(X \geq a)] \text{ since } E[X\mathbb{1}(X < a)] \geq 0 \\
&\geq E[a\mathbb{1}(X \geq a)] \text{ since } X \geq a \\
&= aE[\mathbb{1}(X \geq a)] \\
&= aP(X \geq a)
\end{aligned}$$

where the indicator function is defined as:

$$\mathbb{1}(A) = \begin{cases} 1, & \text{if } A \text{ is true} \\ 0, & \text{otherwise} \end{cases}$$

□

$$\eta = \lim_{n\to\infty} \eta_n = \lim_{n\to\infty} P(S_n = 0) = \lim_{n\to\infty} P(S_n < 1) = 1 - \lim_{n\to\infty} P(S_n \geq 1) \geq 1 - \lim_{n\to\infty} E[S_n]$$

For $\mu < 1$, we have:

$$E[S_n] \to 0 \quad \text{as } n \to \infty$$

$$\Rightarrow \eta \geq 1 \Rightarrow \eta = 1 \text{ ,if } \mu < 1.$$

What happens if $\mu \geq 1$ **and** $\sigma^2 \geq 0$, **with** $\eta_0 = 0$?

$$\begin{aligned}
\eta_n = P(S_n = 0) &= E[P(S_n = 0|S_1)] \\
&= \sum_{k=0}^{\infty} P(S_n = 0|S_1 = k)P(S_1 = k) \\
&= \sum_{k=0}^{\infty} P(S_n = 0|S_1 = k)P(X = k) \\
&= \sum_{k=0}^{\infty} (P(S_{n-1} = 0))^k P(X = k) \\
&= \sum_{k=0}^{\infty} (\eta_{n-1})^k P(X = k) = G_X(\eta_{n-1})
\end{aligned}$$

$$\eta = \lim_{n\to\infty} \eta_n = \lim_{n\to\infty} G_X^{[n]}(0) \text{ recursion repeatedly}$$

also taking limits both sides,

$$\eta = G_X(\eta)$$

$$\therefore \text{ Extinction } = \text{ Finding Roots}$$

Theorem 1.2.7

The probability of ultimate extinction, η, is the smallest non-negative root of

$$\eta = G_X(\eta).$$

Moreover,

$$\eta = \begin{cases} 1, & \mu < 1, \text{ or } \mu = 1 \text{ and } \sigma^2 \geq 0 \\ 0, & \mu \geq 1 \text{ and } \sigma^2 = 0 \\ < 1, & \mu > 1 \text{ and } \sigma^2 > 0. \end{cases}$$

Example.

$X \sim \text{Bern}(p)$ with $p \in [0,1]$. i.e., $P(X=0) = 1-p, \quad P(X=1) = p$.

Answer

$$G(z) = (1-p) + pz.$$

$$\eta = G(\eta) = (1-p) + p\eta.$$

$$(1-p)\eta = (1-p).$$

$$\Rightarrow \eta = \begin{cases} 1, & p \in (0,1) \\ 0, & p \geq 1. \end{cases}$$

Example.

$X \sim \text{Geom}(p)$, where $p \in (0,1)$.

Note:

$$G_X(z) = \frac{p}{1-(1-p)z}, \quad |z| \leq 1.$$

Determine the probability of ultimate extinction, η, as a function of p.

Answer

$$\eta = \frac{p}{1-(1-p)\eta}.$$

$$p = \eta - (1-p)\eta^2.$$

$$0 = (1-p)\eta^2 - \eta + p.$$

$$\eta = \frac{1 \pm \sqrt{1-4p(1-p)}}{2(1-p)}.$$

$$\text{two roots: } \frac{p}{1-p} \text{ and } 1, \text{ since } 1-4p(1-p) = (2p-1)^2.$$

$$\eta = \begin{cases} \frac{p}{1-p}, & p \leq \frac{1}{2} \\ 1, & p > \frac{1}{2}. \end{cases}$$

Alternative way: we know that $\eta = 1$ is a root

$$0 = (\eta - 1)((1-p)\eta - p).$$

$$\Rightarrow \eta = \begin{cases} \frac{p}{1-p}, & p \leq \frac{1}{2} \\ 1, & p > \frac{1}{2}. \end{cases}$$

Example.

> A branching process starts with one individual and the offspring distribution has probability generating function given by
> $$G_X(z) = E[z^X] = \frac{1}{3}(z + z^2 + z^3)$$
> What is the probability of eventual extinction?

$$r = G_X(r) = \frac{1}{3}(r + r^2 + r^3).$$

Multiplying both sides by 3 and rearranging gives

$$0 = r + r^2 + r^3 - 3r = r(r^2 + r - 2),$$

whose real solutions are $r = 0,\ r = 1,$ and $r = -2$. In the interval $[0, 1]$, the only solutions are 0 and 1.

Why $r = 0$ is the correct extinction probability: Because the mean number of offspring $E[X] = G'_X(1)$ exceeds 1 (in fact it is 2 here), the Galton–Watson theory tells us the extinction probability is the *smallest* root in $[0, 1]$. Hence $r = 0$.

Chapter 2

Random Walks

2.1 Introduction to Random Walk

Definition 2.1.1

Let the stochastic process $X = (X_t, t \in \mathcal{T})$ be such that $\forall\, n \in \mathbb{N}$ and $t_0, \dots, t_n \in \mathcal{T}$, $X_{t_0}, \dots, X_{t_n}$ are independent and identically distributed (iid). Consider $\mathcal{T} = \mathbb{N}$ and a new process $S = (S_n, n \in \mathbb{N})$ such that $S_0 = X_0 = x_0$ and $S_n = S_{n-1} + X_n = x_0 + \sum_{k=1}^{n} X_k$. This new "integrated" process is called a Random Walk

Law of large number: If $\mu = E[X]$ exists, then

$$\lim_{n\to\infty} n^{-1} S_n \to \mu \quad \text{(in some sense)}$$

Central limit theory: If $\sigma^2 = \mathrm{Var}(X) < \infty$, then

$$\lim_{n\to\infty} \sigma^{-1} n^{\frac{1}{2}} (n^{-1} S_n - \mu) = \lim_{n\to\infty} \frac{S_n - n\mu}{\sqrt{n}\sigma} \to \mathcal{N}(0,1)(\text{"in distribution"})$$

Example.

What happens for $E[X] = \mu > 0$?

$$E(S_n) = E\left(X_0 + \sum_{i=1}^{n} X_i\right) = x_0 + n\mu$$

Note that the state space for the X-sequence, E_X, is not the same (in general) as the state space for the S-sequence, E_S.

Example.

Say $X_0 = 0$ and $X_1, X_2, \dots \sim \mathrm{Bern}(p)$, where $p \in [0,1]$.
i.e., $P(X_i = 1) = p$, $P(X_i = 0) = 1 - p$, $\forall i$.
What is the distribution of S_n?

Then $\mathbb{P}(X_k = 1) = p = 1 - \mathbb{P}(X_k = 0)$ and $S_n \sim \mathrm{Bin}(n,p)$ with $\mathbb{E}(S_n) = np$ and $\mathrm{Var}(S_n) = np(1-p)$. Similarly, $S_{n+1} \sim \mathrm{Bin}(n+1, p)$ and we have that

$$S_{n+1} \mid S_n \;\sim\; S_n + \mathrm{Ber}(p), \qquad S_{n+k} \mid S_n \;\sim\; S_n + \mathrm{Bin}(k,p), \qquad \forall\, k \in \mathbb{N}_{\geq 1}.$$

Find for $k \in \{0, 1, 2, \dots\}$: $\mathrm{Cov}(S_n, S_{n+k}) = E[S_n S_{n+k}] - E[S_n]E[S_{n+k}]$.

(i)

$$E[S_n S_{n+k}] = E\left[\left(\sum_{i=1}^{n} X_i\right)\left(\sum_{j=1}^{n+k} X_j\right)\right]$$

$$= E\left[\sum_{i=1}^{n}\sum_{j=1}^{n+k} X_i X_j\right]$$

$$= E\left[\sum_{i=1}^{n} X_i^2 + \sum_{i=1}^{n}\sum_{j=1,j\neq i}^{n+k} X_i X_j\right]$$

$$= \sum_{i=1}^{n} E[X_i^2] + \sum_{i=1}^{n}\sum_{j=1,j\neq i}^{n+k} E[X_i X_j]$$

Now,

$$E[X_i^2] = \sigma^2 + \mu^2 = p(1-p) + p^2 = p.$$
$$E[X_i X_j] = E[X_i]E[X_j] = \mu^2 = p^2, \quad \forall i \neq j.$$
$$E[S_n S_{n+k}] = np + n(n+k-1)p^2.$$

(ii)

$$E[S_n]E[S_{n+k}] = np(n+k)p = p^2 n(n+k).$$

Thus,

$$\begin{aligned}\mathrm{Cov}(S_n, S_{n+k}) &= (i) - (ii)\\ &= np(1-p).\end{aligned}$$

$$\mathrm{Corr}(S_n, S_{n+k}) = \frac{\mathrm{Cov}(S_n, S_{n+k})}{\sqrt{\mathrm{Var}(S_n)\mathrm{Var}(S_{n+k})}} = \sqrt{\frac{n}{n+k}}.$$

As k increases, the cor converges to 0.

Example.

> $X_0 = 0, \quad E_X = E_S = \mathbb{R}, \quad X_1, X_2, \cdots \sim$ i.i.d $\mathcal{N}(0, \sigma^2)$.
> Show that
> $$\mathrm{Corr}(S_n, S_{n+k}) = \sqrt{\frac{n}{n+k}}, \quad k = 0, 1, \ldots$$

Find for $k \in \{0, 1, 2, \ldots\}$: $\mathrm{Cov}(S_n, S_{n+k}) = E[S_n S_{n+k}] - E[S_n]E[S_{n+k}]$.

$$E[S_n S_{n+k}] = \sum_{i=1}^{n} E[X_i^2] + \sum_{i=1}^{n}\sum_{j=1,j\neq i}^{n+k} E[X_i X_j]$$

$\mathbb{E}[X_i X_j] = 0$ if $i \neq j$ and $\mathbb{E}[X_i^2] = \sigma^2$. Hence,

$$E[S_n S_{n+k}] \;=\; n\,\sigma^2,$$

Each X_i has mean zero, so

$$\mathbb{E}[S_n] \;=\; \sum_{i=1}^{n}\mathbb{E}[X_i] \;=\; 0, \quad \mathbb{E}[S_{n+k}] \;=\; \sum_{j=1}^{n+k}\mathbb{E}[X_j] \;=\; 0.$$

Thus, $E[S_n]E[S_{n+k}] = 0$. Then

$$\mathrm{Cov}(S_n, S_{n+k}) \;=\; \sum_{i=1}^{n}\sum_{j=1}^{n+k}\mathbb{E}[X_i X_j] \;=\; \sum_{i=1}^{n}\sigma^2 \;=\; n\,\sigma^2.$$

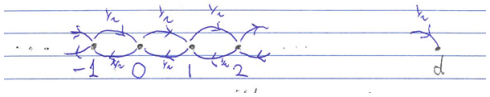
Recall the definition of correlation:

$$\mathrm{Corr}(S_n, S_{n+k}) \;=\; \frac{\mathrm{Cov}(S_n, S_{n+k})}{\sqrt{\mathrm{Var}(S_n)}\,\sqrt{\mathrm{Var}(S_{n+k})}}.$$

Substitute the results from Steps 1 and 2:

$$\mathrm{Corr}(S_n, S_{n+k}) \;=\; \frac{n\,\sigma^2}{\sqrt{n\,\sigma^2}\,\sqrt{(n+k)\,\sigma^2}} \;=\; \sqrt{\frac{n}{n+k}}.$$

2.2 Simple Symmetric Random Walk



Definition 2.2.1: Simple Symmetric Random Walk

Let $X_0 = 0$, and $X_1, X_2, \ldots$ be independent and identically distributed (iid) random variables:

$$X_i \overset{\text{iid}}{\sim} 2\mathrm{Ber}\left(\frac{1}{2}\right) - 1.$$

That is,

$$P(X_i = x) = \begin{cases} \frac{1}{2}, & x = 1, \\ \frac{1}{2}, & x = -1. \end{cases}$$

The position of the walker after n steps is given by:

$$S_n = 0 + \sum_{i=1}^{n} X_i.$$

Definition 2.2.2

S_n : Position of the random walker after n steps.
τ_d : The first time the random walker hits level d, defined as

$$\tau_d = \inf\{n \in \mathbb{N} \mid S_n = d\}.$$

Question: $P(\tau_d < \infty)$?
Define

$$r_k = P(\tau_d < \infty \mid S_0 = k).$$

We know that:

$$r_d = 1.$$

By symmetry, we have:

$$r_{d-i} = r_{d+i}, \quad \forall i \in \mathbb{N}.$$

First-Step Analysis: Consider everything that would happen in one unit of time.

$$r_k = P(\tau_d < \infty \mid S_0 = k)$$

$$= P(\tau_d < \infty \mid S_0 = k, X_1 = 1)P(X_1 = 1 \mid S_0 = k)$$
$$+P(\tau_d < \infty \mid S_0 = k, X_1 = -1)P(X_1 = -1 \mid S_0 = k).$$

Since $P(X_1 = 1) = P(X_1 = -1) = \frac{1}{2}$ and Markov property, this simplifies to:

$$= \frac{1}{2}P(\tau_d < \infty \mid S_1 = k+1) + \frac{1}{2}P(\tau_d < \infty \mid S_1 = k-1).$$

$$= \frac{1}{2}P(\tau_d < \infty \mid S_0 = k+1) + \frac{1}{2}P(\tau_d < \infty \mid S_0 = k-1).$$

$$= \frac{1}{2}r_{k+1} + \frac{1}{2}r_{k-1}.$$

We know:

$$r_d = 1 = \frac{1}{2}(r_{d+1} + r_{d-1}) = \frac{1}{2}(r_{d-1} + r_{d-1}) = r_{d-1}.$$

$$r_{d-1} = \frac{1}{2}(r_d + r_{d-2}) = \frac{1}{2}(r_{d-1} + r_{d-2}).$$

$$\Rightarrow \frac{1}{2}r_{d-1} = \frac{1}{2}r_{d-2}, \quad \text{i.e., } r_{d-1} = r_{d-2}.$$

$$1 = r_d = r_{d-1} = r_{d-2} = \cdots = r_0.$$

Something to note about First-Step Analysis

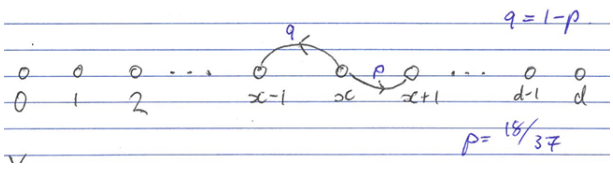
- *Identify the quantity of interest (e.g., τ_d)*
- *We use the following equations:*

$$P(A \mid S_0) = \sum_i P(A \mid S_0, X_i)P(X_i \mid S_0).$$

$$E[\lambda \mid S_0] = E[E[\lambda \mid S_0, X_i]]$$
$$= \sum_i E[A \mid S_0, X_i]P(X_i \mid S_0).$$

- *Set up and solve recursive equations.*

2.3 Simple Random Walk



Definition 2.3.1: Simple Random Walk

Let $S_0 = X_0 = x$, $\quad S_n = x + \sum_{k=1}^{n} X_k$, where $X_1, X_2, \ldots \overset{\text{iid}}{\sim} 2\text{Ber}(p) - 1$. $S_n - S_0$ represents profit/loss.

Example.

$$P(\tau_d < \tau_0 \mid S_0 = x) \equiv P_x(\tau_d < \tau_0) = ?$$

For $0 < x < d$, we use the first-step analysis:

$$u_x = P_x(\tau_d < \tau_0).$$

$$= P_x(\tau_d < \tau_0 \mid X_1 = 1)P(X_1 = 1) + P_x(\tau_d < \tau_0 \mid X_1 = -1)P(X_1 = -1).$$
$$= pP(\tau_d < \tau_0 \mid S_0 = x, S_1 = x + 1) + qP(\tau_d < \tau_0 \mid S_0 = x, S_1 = x - 1).$$
$$= pP(\tau_d < \tau_0 \mid S_0 = x + 1) + qP(\tau_d < \tau_0 \mid S_0 = x - 1).$$
$$= pu_{x+1} + qu_{x-1}.$$

Boundary conditions:

$$u_d = P_d(\tau_d < \tau_0) = 1, \quad u_0 = 0.$$

Since $p+q=1$, we rewrite the equation:

$$(p+q)u_x = pu_{x+1} + qu_{x-1}.$$

Rearranging:

$$p(u_{x+1} - u_x) = q(u_x - u_{x-1}).$$

Define v_x as:

$$v_x = u_x - u_{x-1}.$$

Thus,

$$pv_{x+1} = qv_x.$$

Solving recursively,

$$v_{x+1} = \frac{q}{p}v_x = \left(\frac{q}{p}\right)^2 v_{x-1} = \cdots = \left(\frac{q}{p}\right)^x v_1.$$

From boundary conditions:

$$v_1 = u_1 - u_0 = u_1.$$

$$\therefore v_x = \left(\frac{q}{p}\right)^{x-1} u_1.$$

Since,

$$v_x = u_x - u_{x-1},$$

it follows that:

$$v_{x-1} + v_x = u_x - u_{x-1} - u_{x-2} + u_x - u_{x-1}.$$

$$= u_x - u_{x-2}.$$

$$\sum_{k=1}^{x} v_k = u_x - u_{x-1} + u_{x-2} + \cdots + u_2 + u_1 - u_0 = u_x - u_0 = u_x$$

$$u_x = \sum_{k=1}^{x} \left(\frac{q}{p}\right)^{k-1} u_1 = \sum_{k=0}^{x-1} \left(\frac{q}{p}\right)^{k} u_1 = \begin{cases} u_1 \frac{1-\left(\frac{q}{p}\right)^x}{1-\left(\frac{q}{p}\right)}, & p \neq q \\ u_1 x, & p = q \end{cases}$$

$$1 = u_d = \begin{cases} u_1 \frac{1-\left(\frac{q}{p}\right)^d}{1-\left(\frac{q}{p}\right)}, & p \neq q \\ u_1 d, & p = q = \frac{1}{2} \end{cases}$$

$$u_1 = \begin{cases} \frac{1-\left(\frac{q}{p}\right)}{1-\left(\frac{q}{p}\right)^d}, & p \neq q \\ \frac{1}{d}, & p = q = \frac{1}{2} \end{cases}$$

$$u_x = \begin{cases} \frac{1-\left(\frac{q}{p}\right)^x}{1-\left(\frac{q}{p}\right)^d}, & p \neq q \\ \frac{x}{d}, & p = q = \frac{1}{2} \end{cases}$$

What happens if $d \to \infty$?

$$\text{As } d \to \infty, \quad u_x \to \begin{cases} 1 - \left(\frac{q}{p}\right)^x, & p \geq q \\ 0, & p \leq q \end{cases}$$

Example.

How long will I play before reaching either d or 0?

Let

$$\tau_{\text{od}} = \min(\tau_0, \tau_d)$$

$$h_x = \mathbb{E}\left[\tau_{\text{od}} \mid S_0 = x\right] = \mathbb{E}_x\left[\tau_{\text{od}}\right]$$

$$h_0 = 0, \quad h_d = 0$$

$$h_x = \mathbb{E}_x\left[\tau_{\text{od}} \mid X_1 = 1\right]\mathbb{E}_x\left[X_1 = 1\right] + \mathbb{E}_x\left[\tau_{\text{od}} \mid X_1 = -1\right]\mathbb{E}_x\left[X_1 = -1\right]$$

$$
\begin{aligned}
&= p\mathbb{E}_x\left[\tau_{\mathrm{od}} \mid X_1 = 1\right] + q\mathbb{E}_x\left[\tau_{\mathrm{od}} \mid X_1 = -1\right] \\
&= p\mathbb{E}\left[\tau_{\mathrm{od}} \mid X_1 = 1, S_0 = x\right] + q\mathbb{E}\left[\tau_{\mathrm{od}} \mid X_1 = -1, S_0 = x\right] \\
&= p\mathbb{E}\left[\tau_{\mathrm{od}} \mid S_1 = x+1\right] + q\mathbb{E}\left[\tau_{\mathrm{od}} \mid S_1 = x-1\right]
\end{aligned}
$$

Relabel the time (one time ahead)

$$
\begin{aligned}
&= p\mathbb{E}\left[\tau_{\mathrm{od}} + 1 \mid S_0 = x+1\right] + q\mathbb{E}\left[\tau_{\mathrm{od}} + 1 \mid S_0 = x-1\right] \\
&= ph_{x+1} + qh_{x-1} + 1
\end{aligned}
$$

We now proceed as we did with u_x:

$$
(p+q)\,h_x \;=\; 1 + ph_{x+1} + qh_{x-1}, \qquad p\big(h_{x+1} - h_x\big) = q\big(h_x - h_{x-1}\big) - 1.
$$

$$
\underbrace{h_{x+1} - h_x}_{w_{x+1}} \;=\; \frac{q}{p}\,\underbrace{\big(h_x - h_{x-1}\big)}_{w_x} - \frac{1}{p}, \qquad w_{x+1} = \frac{q}{p}\,w_x - \frac{1}{p}.
$$

$$
w_{x+1} = \frac{q}{p}\,w_x - \frac{1}{p} = \left(\frac{q}{p}\right)^2 w_{x-1} - \frac{q}{p^2} - \frac{1}{p} = \;\dots
$$

$$
w_x = \left(\frac{q}{p}\right)^x w_1 - \frac{1}{p}\Big(1 + \frac{q}{p} + \cdots + \frac{q^{x-1}}{p^{x-1}}\Big) = \left(\frac{q}{p}\right)^x w_1 - \frac{1}{p}\,\frac{1-(q/p)^x}{1-q/p} = \Big(w_1 + \frac{1}{p-q}\Big)\left(\frac{q}{p}\right)^x - \frac{1}{p-q}.
$$

Next, $h_0 = h_b = 0$, so $w_1 = h_1 - h_0 = h_1$ and $w_b = h_b - h_{b-1} = -h_{b-1}$. Furthermore,

$$
w_{x-1} + w_x = h_{x-1} - h_{x-2} + h_x - h_{x-1} = h_x - h_{x-2} \implies \sum_{k=1}^{x} w_k = h_x - h_0 = h_x,
$$

so, using our result for w_x above,

$$
\begin{aligned}
h_x = \sum_{k=1}^{x} w_k &= \sum_{k=1}^{x}\left(\Big(h_1 + \frac{1}{p-q}\Big)\left(\frac{q}{p}\right)^{k-1} - \frac{1}{p-q}\right) \\
&= \Big(h_1 + \frac{1}{p-q}\Big)\sum_{i=0}^{x-1}\left(\frac{q}{p}\right)^i - \frac{x}{p-q} \\
&= \Big(h_1 + \frac{1}{p-q}\Big)\,\frac{1-(q/p)^x}{1-q/p} - \frac{x}{p-q}, \qquad p \neq q, \\
&= h_1 x - \frac{x(x+1)}{2p}, \qquad p = q = \tfrac{1}{2}.
\end{aligned}
$$

To compute the case $p = q = \frac{1}{2}$, start from $w_{x+1} = w_x - \frac{1}{p}$ and continue from there. Hence

$$
h_b = 0 = \Big(h_1 + \frac{1}{p-q}\Big)\,\frac{1-(q/p)^b}{1-q/p} - \frac{b}{p-q}, \qquad p \neq q, \qquad \text{or} \qquad h_b - \frac{b(b+1)}{2p}, \quad p = q = \tfrac{1}{2},
$$

which gives

$$
h_1 = \begin{cases} \dfrac{b}{p-q}\,\dfrac{1-q/p}{1-(q/p)^b} - \dfrac{1}{p-q}, & p \neq q, \\[2ex] \dfrac{b+1}{2p}, & p = q = \tfrac{1}{2}, \end{cases}
$$

meaning

$$
h_x = \begin{cases} \dfrac{b}{p-q}\,\dfrac{1-(q/p)^x}{1-(q/p)^b} - \dfrac{x}{p-q}, & p \neq q, \\[2ex] \dfrac{b\,x - x^2}{2p}, & p = q = \tfrac{1}{2}. \end{cases}
$$

Chapter 3

Discrete Time Markov Chains

3.1 Stochastic process

Definition 3.1.1: Stochastic process

A stochastic process $X(t), t \in T$ is a collection of random variables (vectors) indexed by indexed by some index set T.

- T: totally ordered set
- $s, t \in T$, where $s \leq t$ makes sense.

We specify the *probabilistic* behaviour of a stochastic process X via its finite–dimensional distributions (FDDs).

Definition 3.1.2: Finite Dimensional Distributions (FDDs)

For any $n \in \mathbb{Z}_{>0}$ and $t_1, \ldots, t_n \in \mathcal{T}$, we denote such a probability by

$$\mathbb{P}\big(X(t_1) \leq x_1, \ldots, X(t_n) \leq x_n\big), \qquad X(t) \in \Omega \subseteq \mathbb{R}^d,$$

where Ω is the state space of X and $d \in \mathbb{Z}_{>0}$.

3.2 Introduction to DTMC

Definition 3.2.1: Markov Process

Let $(X_t, t \in \mathcal{T})$ be a stochastic process. If, for any $s, t \in \mathcal{T}$ and any $A \subseteq \Omega$ (state space),

$$\mathbb{P}\big(X_{t+s} \in A \mid \mathcal{F}_t\big) = \mathbb{P}\big(X_{t+s} \in A \mid X_t\big),$$

then $(X_t, t \in \mathcal{T})$ is called a *Markov process*, where $\mathcal{F}_t$ is the "history of the process up to and including time t". The above property is known as the *Markov property*.

Definition 3.2.2: Discrete-Time Markov Chain (DTMC)

A sequence $(X_n, n = 0, 1, \dots)$ is a DTMC if the Markov property holds, that is,

$$P(X_{n+1} = j \mid X_0 = x_0, \dots, X_{n-1} = x_{n-1}, X_n = i) = P(X_{n+1} = j \mid X_n = i),$$

for all $n \in \mathbb{N}$, and all $i, j, x_0, \dots, x_{n-1} \in E$.
This formulation appears less general.

Claim

We can obtain all probabilities of the form

$$P_{ij}^{(m)}(n) = P(X_{n+m} = j \mid X_n = i)$$

from

$$P_{ij}^{(1)}(n) = P(X_{n+1} = j \mid X_n = i), \quad \forall i, j \in E, \quad n \in \mathbb{N}.$$

Proof for Claim.

We see that

$$\begin{aligned}
P_{ij}^{(2)}(n) &= \mathbb{P}(X_{n+2} = j \mid X_n = i) \\
&= \sum_{k \in \Omega} \mathbb{P}(X_{n+2} = j \mid X_n = i, X_{n+1} = k)\, \mathbb{P}(X_{n+1} = k \mid X_n = i) \\
&= \sum_{k \in \Omega} \mathbb{P}(X_{n+2} = j \mid X_{n+1} = k)\, \mathbb{P}(X_{n+1} = k \mid X_n = i) \\
&= \sum_{k \in \Omega} P_{kj}^{(1)}(n+1)\, P_{ik}^{(1)}(n) \\
&= \sum_{k \in \Omega} P_{ik}^{(1)}(n)\, P_{kj}^{(1)}(n+1).
\end{aligned}$$

Hence, we see that

$$P^{(1)}(n) = \big(P_{ij}^{(1)}(n),\, i, j \in \Omega\big), \qquad P^{(2)}(n) = \big(P^{(1)}(n) P^{(1)}(n+1)\big)_{ij}, \qquad P^{(m)}(n) = \big(P_{ij}^{(1)}(n),\, i, j \in \Omega\big),$$

meaning

$$P^{(m)}(n) = P^{(1)}(n)\, P^{(1)}(n+1) \cdots P^{(1)}(n+m-1).$$

Example.

Consider a Markov Chain (MC) with $E = \{0, 1\}$. Rransition probability matrices:

$$P^{(1)}(n) = \begin{bmatrix} 0.8 & 0.2 \\ 0.5 & 0.5 \end{bmatrix}$$

i.e. $P(X_{n+1} = 1 \mid X_n = 0) = 0.2$.

$$P^{(1)}(n+1) = \begin{bmatrix} 0.6 & 0.4 \\ 0.3 & 0.7 \end{bmatrix}$$

$$P^{(2)}(n) = \begin{bmatrix} 0.8 & 0.2 \\ 0.5 & 0.5 \end{bmatrix} \begin{bmatrix} 0.6 & 0.4 \\ 0.3 & 0.7 \end{bmatrix}$$

Computing the matrix multiplication:

$$= \begin{bmatrix} 0.8 \times 0.6 + 0.2 \times 0.3 & 0.8 \times 0.4 + 0.2 \times 0.7 \\ 0.5 \times 0.6 + 0.5 \times 0.3 & 0.5 \times 0.4 + 0.5 \times 0.7 \end{bmatrix}$$

$$= \begin{bmatrix} 0.54 & 0.46 \\ 0.45 & 0.55 \end{bmatrix}$$

We now want to be able to specify $\mathbb{P}\big(X_{i_0} = x_0, \dots, X_{i_n} = x_n\big)$ *for any* $n \in \mathbb{N}$, $i_0 < \dots < i_n \in \mathcal{T}$ *and* $x_0, \dots, x_n \in \Omega$. *To do so, we need* $P^{(m)}(n)$ *for all* $m, n \in \mathbb{N}$ *and* $\pi_x^{(0)} = \mathbb{P}(X_0 = x)$ *for all* $x \in \Omega$. *Define* $\underline{\pi}^{(0)} := \big(\pi_x^{(0)}, x \in \Omega\big)$ *as a row vector. Then, we see that*

$$\mathbb{P}\big(X_{i_0} = x_0, \dots, X_{i_n} = x_n\big) = \sum_{x \in \Omega} \pi_x^{(0)} P_{x,x_0}^{(i_0)}(0) \prod_{k=1}^{n} P_{x_{k-1},x_k}^{(i_k - i_{k-1})}(i_{k-1}).$$

3.3 Limiting Distribution

3.3.1 Time–homogeneous DTMC

Definition 3.3.1

Let $(X_t, t \in \mathcal{T})$ be a Markov process. If $P^{(1)}(n)$ does *not* depend on n for all $n \in \mathbb{N}$, the process is called *time homogeneous*. In such a case, we denote $P^{(1)}(n)$ by P and $P^{(m)}$ by P^m.

For a time–homogeneous DTMC,

$$P^{(1)}(n) = P \quad \forall n$$

$$P^{(m)}(n) = P^{(1)}(n)P^{(1)}(n+1)\dots P^{(1)}(n+m-1) = P^m$$

set $\pi_x^{(m)} = \mathbb{P}(X_m = x)$ *and collect these into the row vector* $\underline{\pi}^{(m)} = \big(\pi_x^{(m)}, x \in \Omega\big)$. *Then* $\underline{\pi}^{(m)} = \underline{\pi}^{(0)} P^m$, *where* $\underline{\pi}^{(0)}$ *is the initial distribution of the* X_t *and* P *is the one–step transition matrix. It is easy to see that we require*

$$\sum_{j \in \Omega} P_{ij}^{(1)}(n) = 1, \qquad \sum_{x \in \Omega} \pi_x^{(0)} = 1.$$

	$\boldsymbol{\pi}^{(0)}$	P
$\lvert\Omega\rvert < \infty$	finite–dim. vector	matrix
$\lvert\Omega\rvert = \infty$	infinite–dim. vector	linear operator

Theorem 3.3.2: Chapman–Kolmogorov equations

Let $(X_t, t \in \mathcal{T})$ be a Markov process. Then $P^{(m+n)} = P^{(m)}P^{(n)}$ for all $m, n \in \mathbb{N}$ (if the process is time homogeneous, we have $P^{m+n} = P^m P^n$).

3.3.2 Finding Limiting Distribution

The transition matrix P is always square. If $|E| = r < \infty$ and all the eigenvectors of P, $\lambda_1, \dots, \lambda_r$ are **distinct**, then P is diagonalizable, i.e., $P = V\Lambda V^{-1}$ where

$$\Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_r) = \begin{bmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_r \end{bmatrix} \Rightarrow \Lambda^n = \begin{bmatrix} \lambda_1^n & 0 \\ 0 & \lambda_r^n \end{bmatrix}$$

$$P^2 = V\Lambda V^{-1} V\Lambda V^{-1} = V\Lambda^2 V^{-1} \Rightarrow P^n = V\Lambda^n V^{-1}$$

Want $P = V\Lambda W \quad \Rightarrow$ Show we can construct using eigenvectors and eigenvalues.

Definition 3.3.3

Right Eigenvectors v_i: (column) $P\underline{v_i} = \lambda_i \underline{v_i}$, where λ_i is an eigenvalue.
Left Eigenvectors w_i: (row) $\underline{w_i}P = \lambda_i \underline{w_i}$, where λ_i is an eigenvalue.
Require (normalized): $\underline{w_i v_i} = 1$, $i = 1, \dots, r$.

Proposition 3.3.4

Let P have distinct eigenvalues $\lambda_1, \dots, \lambda_r$, and let $\underline{v_j}$ be the jth right eigenvector, $\underline{w_i}$ be the ith left eigenvector.
Then,

$$\underline{w_i}\underline{v_j} = 0, \quad \forall i \neq j.$$

Proof. We have $\underline{w}_j P = \lambda_j \underline{w}_j$ and $P\underline{v}_i = \lambda_i \underline{v}_i$. Then

$$\underline{w}_j P \underline{v}_i = \lambda_j \underline{w}_j \underline{v}_i.$$

$$\lambda_j \underline{w}_j \underline{v}_i = \lambda_i \underline{w}_j \underline{v}_i.$$

$$(\lambda_j - \lambda_i)\underline{w}_j \underline{v}_i = 0.$$

Since λ_j and λ_i are distinct,

$$\underline{w}_j \underline{v}_i = 0.$$

□

Claim

$W = V^{-1}$

Proof for Claim.

$$\underline{WV} = \begin{bmatrix} \underline{w}_1 \\ \vdots \\ \underline{w}_r \end{bmatrix} \begin{bmatrix} \underline{v}_1 & \dots & \underline{v}_r \end{bmatrix} = \begin{bmatrix} \underline{w}_1\underline{v}_1 & \underline{w}_1\underline{v}_2 & \dots \\ \underline{w}_2\underline{v}_1 & \underline{w}_2\underline{v}_2 & \dots \\ \vdots & \vdots & \ddots \end{bmatrix} = \begin{bmatrix} 1 & 0 & \dots \\ 0 & 1 & \dots \\ \vdots & \vdots & \ddots \end{bmatrix} = I.$$

$$P^n = \underline{V}\Lambda^n \underline{W} = \begin{bmatrix} \underline{v}_1 & \dots & \underline{v}_r \end{bmatrix} \begin{bmatrix} \lambda_1^n & 0 & \dots \\ 0 & \lambda_2^n & \dots \\ \vdots & \vdots & \ddots \end{bmatrix} \begin{bmatrix} \underline{w}_1 \\ \vdots \\ \underline{w}_r \end{bmatrix}$$

$$= \lambda_1^n \underline{v}_1 \underline{w}_1 + \lambda_2^n \underline{v}_2 \underline{w}_2 + \dots + \lambda_r^n \underline{v}_r \underline{w}_r$$

$$= \lambda_1^n R_1 + \lambda_2^n R_2 + \dots + \lambda_r^n R_r.$$

$$\text{where} \quad R_i = \underline{v_i}\,\underline{r_i} == \begin{pmatrix} v_{i1} \\ v_{i2} \\ \vdots \\ v_{ir} \end{pmatrix} \begin{pmatrix} w_{i1} & w_{i2} & \dots & w_{ir} \end{pmatrix} = \begin{pmatrix} v_{i1}w_{i1} & v_{i1}w_{i2} & \dots & v_{i1}w_{ir} \\ v_{i2}w_{i1} & v_{i2}w_{i2} & \dots & v_{i2}w_{ir} \\ \vdots & \vdots & \ddots & \vdots \\ v_{ir}w_{i1} & v_{ir}w_{i2} & \dots & v_{ir}w_{ir} \end{pmatrix}$$

Theorem 3.3.5

Properties of R_i:

- $R_i R_i = \underline{v_i w_i v_i w_i} = 1\underline{v_i w_i} = R_i$
- $R_i R_j = \underline{v_i w_i v_j w_j} = 0$
- For any DTMC, $\underline{1}$ is a right eigenvector

$$P^n = 1^n R_1 + \lambda_2^n R_2 + \cdots + \lambda_r^n R_r = \begin{pmatrix} \underline{w} \\ \underline{w} \\ \vdots \\ \underline{w} \end{pmatrix} + \lambda_2^n R_2 + \cdots + \lambda_r^n R_r$$

Denote $\underline{w}$ by $\underline{\pi}$. If further we have $|\lambda_k| < 1, \quad k = 2, 3, \ldots, r$, then

$$\lim_{n\to\infty} P^n = \begin{pmatrix} \underline{\pi} \\ \vdots \\ \underline{\pi} \end{pmatrix}$$

and we call π the **limiting distribution**.

Theorem 3.3.6: Gershgorin Circle Theorem – adapted

Let P be the transition matrix for a Markov process and let D_i be closed discs in $\mathbb{C}$ with centres P_{ii} and radii $r_i = 1 - P_{ii}$. Then every eigenvalue λ_i of P lies in at least one of $\{D_i\}$.

Definition 3.3.7: Limiting Distribution

Recall $(P^n)_{ij} = P(X_n = j \mid X_0 = i)$ and define $\forall i, j \in E$,$\lim_{n\to\infty} P(X_n = j \mid X_0 = i) = P_{ij}^{(\infty)}$. If it exists and $P_{ij}^{(\infty)} = p_j^{(\infty)} \quad \forall i, j \in E$, then $\underline{p}^{(\infty)} = (p_j^{(\infty)}, j \in E)$ is the limiting distribution.

Definition 3.3.8: Stationary Distribution

If there is a distribution $\underline{\pi}$ on E that satisfies $\underline{\pi}P = \underline{\pi}$, then $\underline{\pi}$ is a *stationary distribution* for P.

If $\underline{\pi}^{(0)} = \underline{\pi}$, then

$$\underline{\pi}^{(1)} = \underline{\pi}^{(0)} P = \underline{\pi} P = \underline{\pi}.$$

For our limiting distribution,

$$\underline{p}^{(\infty)} = \lim_{n\to\infty} \underline{p}^{(n)} = \lim_{n\to\infty} \underline{p}^{(n+1)} = \lim_{n\to\infty} \underline{p}^{(n)} P = \underline{p}^{(\infty)} P.$$

It is stationary.

Example.

$$P = \begin{bmatrix} 0 & \frac{1}{3} & \frac{2}{3} \\ 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$$

Solve $\underline{\pi}P = \underline{\pi}$. We know $\underline{\pi}1 = 1$.

$$(\pi_0 \ \pi_1 \ \pi_2) \begin{pmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix} = (\pi_0 \ \pi_1 \ \pi_2),$$

which yields

$$\frac{1}{3}\pi_0 + \frac{1}{2}\pi_2 = \pi_0, \qquad \frac{1}{3}\pi_0 + \frac{1}{2}\pi_2 = \pi_1, \qquad \pi_0 + \pi_1 + \pi_2 = 1.$$

Hence $\pi_2 = \frac{4}{3}\pi_0$. Plugging back gives $\frac{1}{3}\pi_0 + \frac{2}{3}\pi_0 = \pi_1 \implies \pi_0 = \pi_1$. Using the normalisation,

$$\pi_0 + \pi_0 + \tfrac{4}{3}\pi_0 = 1 \implies \pi_0 = \tfrac{3}{10}, \quad \pi_1 = \tfrac{3}{10}, \quad \pi_2 = \tfrac{4}{10}.$$

3.3.3 Represent the Transition Matrix

There are a few ways we can represent the transition matrix. For instance, consider a system in which the state space is $\Omega = \{0, 1, 2, 3, 4, 5\}$ and the "equation form" of P is

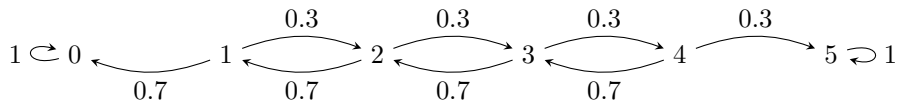
$$P_{00} = 1, \qquad P_{55} = 1, \qquad P_{i,i+1} = 0.3 \ \ \forall i \in \{1, 2, 3, 4\}, \qquad P_{i,i-1} = 0.7 \ \ \forall i \in \{1, 2, 3, 4\}.$$

If only a few elements in the state space interact (and the state space is small), this is probably the most readable way to present P. Naturally, there is the usual matrix form:

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0.7 & 0 & 0.3 & 0 & 0 & 0 \\ 0 & 0.7 & 0 & 0.3 & 0 & 0 \\ 0 & 0 & 0.7 & 0 & 0.3 & 0 \\ 0 & 0 & 0 & 0.7 & 0 & 0.3 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix},$$

where the columns and rows are both numbered 0 to 5 starting from the top-left corner; each matrix element is the transition probability from the row index to the column index.

Finally, this same information can be represented in a *transition probability graph*:



Fact 3.3.9

Algorithm 1 (Simulating from a DTMC): We have $E = \{0, 1, 2, \dots\}$ and $P = \begin{pmatrix} \underline{p_0} \\ \underline{p_1} \\ \underline{p_2} \\ \vdots \end{pmatrix}$

1. $X_0 \sim \underline{\pi}^{(0)}$.
2. For $n = 1, 2, \dots$
3. $X_n \sim \underline{p}_{X_{n-1}}$.

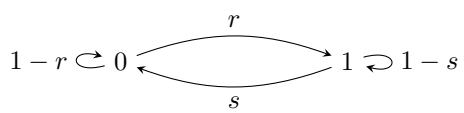
3.3.4 Example

Consider the DTMC with state space

$$\Omega = \{0, 1\}, \qquad \text{initial distribution } \boldsymbol{\pi}^{(0)} = (1 - p, p) = (q, p) \text{ with } p \in [0, 1],$$

and transition matrix

$$P = \begin{pmatrix} 1 - r & r \\ s & 1 - s \end{pmatrix}, \qquad r, s \in [0, 1].$$



P has two eigenvalues, λ_1 and λ_2, of which we let $\lambda_1 = 1$ (recall that 1 must always be an eigenvalue). Recall also, from linear algebra, that

$$\operatorname{tr}(P) = \sum_k \lambda_k \quad \text{and} \quad \det(P) = \prod_k \lambda_k.$$

Thus, we see that

$$\operatorname{tr}(P) = 1 - r + 1 - s = 1 + \lambda_2 \implies \lambda_2 = 1 - r - s.$$

There are now two cases to consider.

First, if $\lambda_2 = 1$. Then $r = s = 0$, so $P = I$ and we have $\boldsymbol{\pi}^{(n)} = \boldsymbol{\pi}^{(0)} P^n = \boldsymbol{\pi}^{(0)} = (q, p)$, meaning every distribution is stationary.

Second, if at least one of r and s is not 0. Then $\lambda_1 = 1$ and $\lambda_2 = 1 - r - s \neq 1$. We want to write

$$P^n = \lambda_1^n R_1 + \lambda_2^n R_2 = R_1 + (1 - r - s)^n R_2.$$

To find R_1 and R_2, consider the cases $n = 0$ and $n = 1$:

$$P^0 = I = R_1 + R_2 \implies R_1 = I - R_2,$$

$$P = R_1 + (1 - r - s)R_2 = I - R_2 + (1 - r - s)R_2.$$

Hence, we have

$$R_2 = \frac{1}{r+s}(I - P) = \frac{1}{r+s}\begin{pmatrix} r & -r \\ -s & s \end{pmatrix} \quad \implies \quad R_1 = I - R_2 = \frac{1}{r+s}\begin{pmatrix} s & r \\ s & r \end{pmatrix},$$

and consequently,

$$P^n = \frac{1}{r+s}\begin{pmatrix} s & r \\ s & r \end{pmatrix} + \big(1 - (r+s)\big)^n \frac{1}{r+s}\begin{pmatrix} r & -r \\ -s & s \end{pmatrix}.$$

There are now two options:

1. $|1 - r - s| < 1$. In this case,

$$\lim_{n\to\infty} P^n = \frac{1}{r+s}\begin{pmatrix} s & r \\ s & r \end{pmatrix},$$

meaning there exists a *unique* limiting distribution (which we know is also stationary). We find it in the usual way by solving $\boldsymbol{\pi} P = \boldsymbol{\pi}$:

$$(\pi_0 \ \pi_1)\begin{pmatrix} 1-r & r \\ s & 1-s \end{pmatrix} = (\pi_0 \ \pi_1).$$

This leads to $(1 - r)\pi_0 + s\pi_1 = \pi_0$ and $\pi_0 + \pi_1 = 1$, hence $\pi_1 = 1 - \pi_0$. Thus,

$$(1 - r)\pi_0 + s(1 - \pi_0) = \pi_0 \implies \pi_0 = \frac{s}{r+s} \implies \pi_1 = \frac{r}{r+s}.$$

2. $1 - (r + s) = -1$. In this case, both $r = s = 1$, meaning our state transitions from 0 to 1 or 1 to 0 at every step. Here,

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \implies P^{2m+1} = P, \quad P^{2m} = I, \qquad \forall m \in \mathbb{N}.$$

It follows that there is no limiting distribution. However, $(q, p)P = (p, q)$, meaning there is a unique *stationary* distribution, namely $\boldsymbol{\pi} = (1/2, 1/2)$.

3.4 Communication classes

Let $(X_n, n = 0, 1, \dots)$ be a Markov chain. Define,

$$P_x(\cdot) := P(\cdot | X_0 = x) \quad E_x(\cdot) := E[\cdot | X_0 = x]$$

$$\tau_y = \inf\{n \geq 0; X_n = y\} \quad \tau_y^{(k)} = \inf\{n \geq \tau_y^{(k-1)}; X_n = y\}$$

$$r_{yy}^{(k)} = P_y(\tau_y^{(k)} < \infty) \quad r_{xy}^{(k)} = P_x(\tau_y^{(k)} < \infty)$$

Example.

$$r_{yy}^{(2)} = P_y(\tau_y^{(2)} < \infty) = P(\tau_y^{(2)} < \infty | \tau_y^{(1)} < \infty) P_y(\tau_y^{(1)} < \infty)$$

Since $\tau_y^{(1)}, \tau_y^{(k+1)} - \tau_y^{(k)} | X_0 = y$ are i.i.d., $= P_y(\tau_y^{(1)} < \infty) r_{yy} = r_{yy}^2$.

Definition 3.4.1

If $r_{yy} < 1$, so that $r_{yy}^k \to 0 \quad$ as $k \to \infty$, then y is a **transient state**.
If $r_{yy} = 1$, so that $r_{yy}^k = 1$ for all $k = 0, 1, \ldots$, then call y a **recurrent state**.

Definition 3.4.2

If $r_{xy} = P_x(\tau_y < \infty) > 0$, we say x **leads to** y and write $x \to y$.
If $r_{xy} = P_x(\tau_y < \infty) = 0$, we say x does not **lead to** y and write $x \not\to y$.

Definition 3.4.3

If $x \to y$ and $y \to x$, then we say x and y **communicate** and write: $x \leftrightarrow y$.

*This is a **symmetric** and **binary relation**.*

Lemma 3.4.4

1. If $x \to y$ and $y \to z$ then $x \to z$.
2. If $x \leftrightarrow y$ and $y \leftrightarrow z$ then $x \leftrightarrow z$.

Proof. (i) If $x \to y$, then $r_{xy} > 0$.

$$\Rightarrow \exists m \geq 1 \text{ such that } P_x(X_m = y) > 0.$$

Similarly,

$$\exists n \geq 1 \text{ such that } P_y(X_n = z) > 0.$$

$$P_x(X_{n+m} = z) = \sum_{\gamma \in E} P(X_{n+m} = z \mid X_n = \gamma) P_x(X_n = \gamma)$$

$$\geq P_y(X_{m+n} = z | X_n = y) \cdot P_x(X_n = y) = P_y(X_m = z) \cdot P_x(X_n = y) = r_{yz} \cdot r_{xy} > 0.$$

$$\therefore r_{xz} > 0$$

$$\Rightarrow x \to z.$$

(ii) We have $x \to y$, $y \to z$ from (i), so $x \to z$. We also have $x \leftrightarrow y$ and $y \leftrightarrow z$, which implies $x \leftrightarrow z$. □

Lemma 3.4.5

If $x \to y$ but $r_{yx} < 1$, then x is transient.

Lemma 3.4.6

If x is recurrent and $x \to y$, then y is recurrent and $y \to x$.

Proof. If $\gamma_{yx} < 1$, then Lemma implies x is transient, so we must have $\gamma_{yx} = 1$. Then $\gamma_{yx} > 0$, so $y \to x$. y is recurrent because $\gamma_{xx} = 1 \implies \gamma_{xx}^{(m)} = 1$ for all m, so we must have $\gamma_{yy} = 1$. □

If $x \leftrightarrow x$, then $\leftrightarrow$ is symmetric, transitive, and reflexive, i.e., $\leftrightarrow$ is an **equivalence relation**. We will use this to construct equivalence classes of states which are in mutual communication, i.e., **communicating classes**.

Definition 3.4.7

A set of states $A \subseteq E$ is called a **communicating class** if $\forall x, y \in A$, $x \leftrightarrow y$ and any $z \in A^c$ there is no $x \in A$ such that $x \leftrightarrow z$.

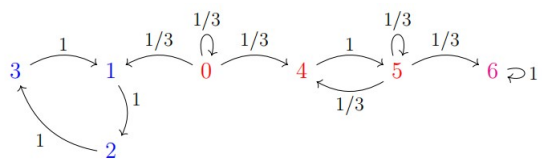
Definition 3.4.8

If $A = E$, the state space is called **irreducible**. A set of states $A \subseteq E$ is **closed** if $r_{xy} = 0 \; \forall x \in A$ and any $y \in A^c$. If $A = \{a\}$ then state a is called **absorbing**.

Note that closed and recurrent classes are equivalent for a finite state space. The distinction is only relevant for infinite state spaces.

Example.

Determine the Communicating Classes



In this case, $T = \{0, 4, 5\}$, $R_1 = \{1, 2, 3\}$ and $R_2 = \{6\}$. In terms of the matrix, the red section corresponds to T_0, green to T_1, orange to T_2, blue to R_1 and magenta to R_2.

If $|E| < \infty$, it can be partitioned into a disjoint union of sets of states:

$$E = T \cup R_1 \cup R_2 \cup \cdots \cup R_k.$$

where:

- T: transient states
- $R_1, \ldots, R_k$: communicating classes of recurrent states

Rearranging the Transition Matrix:
We can always relabel states in P such that:

$$P = \begin{bmatrix} T_0 & T_1 & T_2 & \ldots & T_k \\ 0 & R_1 & 0 & \ldots & 0 \\ 0 & 0 & R_2 & \ldots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \ldots & R_k \end{bmatrix}$$

Then the transition matrix could look like

$$\begin{pmatrix} 1/3 & 1/3 & 0 & 1/3 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1/3 & 1/2 & 0 & 0 & 0 & 1/3 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix},$$

Lemma 3.4.9

If $C \subseteq E$ is a finite closed communicating class, then all $x \in C$ are recurrent.

Definition 3.4.10

Think about $p_{xx}^n > 0$ or $p_{xx}^n = 0$. (n means steps)

$$I_x = \{n > 1 : p_{xx}^n > 0\}$$

Call $\delta_x = \gcd(I_x)$ the period of state x. If $\delta_x = 1$, then we say x is **aperiodic**.

Lemma 3.4.11

If $x \leftrightarrow y$, then x and y have the same period. Moreover, if $p_{xx} > 0$, then x is aperiodic.

$$\Rightarrow \text{ All states in a communicating class have the same period.}$$

Definition 3.4.12

If $\mathbb{E}_x[\tau_x] < \infty$, then x is called *positive recurrent*. If $\mathbb{E}_x[\tau_x] = +\infty$, the state x is *null recurrent*.

Theorem 3.4.13

A finite Markov chain with recurrent communicating class will always be *positive recurrent*.

Theorem 3.4.14

- If X is irreducible and positive recurrent, then there is a unique stationary distribution $\boldsymbol{\pi}$.
- Moreover, if X is also aperiodic, then

$$\lim_{n\to\infty} \underline{\pi}^{(n)} = \underline{\pi}.$$

(this Markov chain is called *ergodic*)

Finally, for an irreducible Markov chain, the following are equivalent:

(i) There is a stationary distribution $\underline{\pi}$.

(ii) Some state $x \in E$ is positive recurrent.

(iii) All states in E are positive recurrent.

3.5 Global and local balance

Definition 3.5.1

Given a Markov process with transition matrix P and state space Ω, its stationary distribution satisfies $\pi P = \pi$ and $\pi \mathbf{1} = 1$. This means the stationary distribution must satisfy the **global balance equations**:

$$\sum_{x\in\Omega} \pi_x P_{xy} = \pi_y, \quad \forall y \in \Omega.$$

If $\{\pi_x\}_{x\in\Omega}$ satisfies the global balance equations and $\sum_{x\in\Omega} \pi_x = 1$, we say $\{\pi_x\}_{x\in\Omega}$ is stationary for the Markov chain.

Definition 3.5.2

There is a similar condition known as the local balance equations: $\pi_x P_{xy} = \pi_y P_{yx} \quad \forall x, y \in E$.

Claim

If $\{\pi_x, x \in E\}$ satisfies local balance, then it also satisfies global balance.

Proof for Claim.

$$\sum_{x \in E} \pi_x P_{xy} = \sum_{x \in E} \pi_y P_{yx} = \pi_y \sum_{x \in E} P_{yx} = \pi_y \cdot 1 = \pi_y \quad \forall y.$$

Definition 3.5.3

Suppose $|E| < \infty$ and $\sum_{y \in E} P_{xy} = 1 \ \forall x \in E$ and $\sum_{x \in E} P_{xy} = 1 \ \forall y \in E$. Then P is called **doubly stochastic**.

Lemma 3.5.4

If $|E| < \infty$ and P is doubly stochastic, then $\underline{\pi} = \frac{1}{|E|}(1, 1, \ldots, 1)$ is stationary for P.

Proof.

$$\text{WTS } \pi_y = \sum_{x \in E} \pi_x P_{xy} \quad \forall y \in E$$

$$\pi_y = \sum_{x \in E} \frac{1}{|E|} P_{xy} = \frac{1}{|E|} \sum_{x \in E} P_{xy} = \frac{1}{|E|}$$

□

Example.

Consider the Markov process with state space $\Omega = \{0, 1, 2\}$ and transition matrix

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}.$$

We see that P is doubly stochastic, $\pi = (1/3, 1/3, 1/3)$ is stationary for P. Indeed,

$$\begin{pmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{pmatrix}.$$

Example.

Find a stationary distribution for this chain.

$$P = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 0 & 1/2 & 1/2 \\ 1/2 & 0 & 1/2 \end{bmatrix}$$

$$\begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} 1/2 & 1/2 & 0 \\ 0 & 1/2 & 1/2 \\ 1/2 & 0 & 1/2 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix}$$

Definition 3.5.5

Undirected simple graph with adjacency matrix $A = (A_{ij}),\ i, j \in \{1, \dots, v\}$

$$A_{ij} = \begin{cases} 1 & \text{if } (i,j) \text{ is an edge} \\ 0 & \text{otherwise} \end{cases}$$

$$A_{ij} = A_{ji}, \quad A = A^\top$$

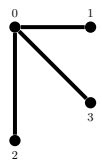
Definition 3.5.6

Suppose (A_{ij}) is the adjacency matrix of a simple undirected graph. Define:

$$P_{ij} = \frac{A_{ij}}{d_i}, \quad d_i = \deg(i), \text{ where d is the degree of neighborhood}$$

Then we call the Markov chain with 1-step transition matrix P a random walk on the graph encoded in (A_{ij}).

Example.



Below is the 1-step transition matrix of the above undirected simple graph.

Then

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}, P = \begin{bmatrix} 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

Claim

$\pi_i = cd_i$, where $c = \left(\sum_i d_i\right)^{-1}$ is stationary for a random walk on a simple undirected graph with degree sequence $(d_i, i \in \Omega)$.

Proof for Claim.

Recall: Local balance

$$\pi_j P_{ji} = \pi_i P_{ij} \quad \forall i, j \in E$$

$$LHS = \pi_j P_{ji} = cd_j \cdot \frac{A_{ji}}{d_j} = cA_{ji} = cd_i \cdot \frac{A_{ij}}{d_i} = \pi_i P_{ij} = RHS$$

Example.

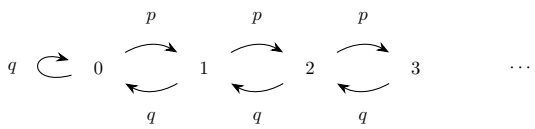
For previous: $\underline{\pi} = \frac{1}{6}(3,\ 1,\ 1,\ 1)$ is a stationary distribution.

Example.

Suppose $(X_n, n \in \mathbb{N})$ is a Markov chain with state space $\Omega = \mathbb{N}$ and

$$P_{x,x+1} = p \quad x \in \mathbb{N}, \quad P_{x,x-1} = 1 - p = q \quad x \in \mathbb{N}_{\geq 1}, \quad P_{0,0} = q.$$

The corresponding transition probability graph is shown below.



Now observe that for $y = x + 1 \in \Omega$, we have

$$\pi_x P_{x,x+1} = \pi_{x+1} P_{x+1,x}, \quad x \in \mathbb{N}.$$

Hence,

$$\pi_x p = \pi_{x+1} q \quad \Rightarrow \quad \pi_{x+1} = \frac{p}{q}\pi_x = \left(\frac{p}{q}\right)^2 \pi_{x-1} = \cdots = \left(\frac{p}{q}\right)^x \pi_0.$$

Recall that $\sum_{x \geq 0} \pi_x = 1$. In this case, assuming $\pi_0 \in (0, \infty)$ and letting $r = p/q$, we have

$$\sum_{x=0}^{\infty} \left(\frac{p}{q}\right)^x \pi_0 = \pi_0 \sum_{x=0}^{\infty} r^x = \begin{cases} \frac{\pi_0}{1-r}, & r \in [0,1), \\ \infty, & r \geq 1. \end{cases}$$

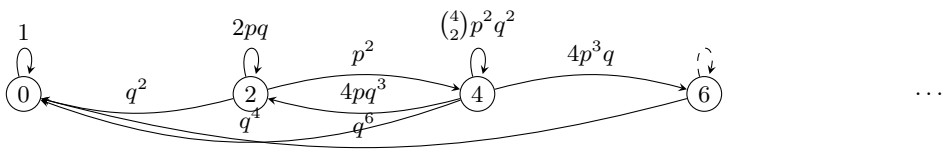
Thus, if $r \geq 1$ $(p \geq q)$, there is no stationary distribution from local balance. Meanwhile, if $0 \leq r < 1$ $(0 \leq p < q)$, then $\pi_0 = 1 - r$ and

$$\pi_x = (1-r)r^x = \left(1 - \frac{p}{q}\right)\left(\frac{p}{q}\right)^x, \quad x \in \mathbb{N} \Rightarrow \text{Geometric starting at 0 stationary distribution}$$

We can represent any branching process as a Markov chain in this way.

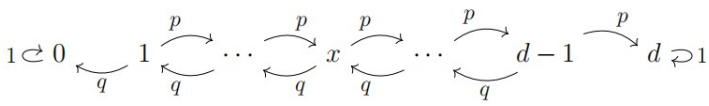
Example.

Say $(S_n, n = 0, 1, \ldots)$ is a branching process with $S_0 = 1$ and offspring distribution $\mathbb{P}(X = 2) = p$ and $\mathbb{P}(X = 0) = 1 - p = q$.



Example.

the transition probability graph for the Gambler's Ruin branching process (with $\pi_x^{(0)} = 1$)



3.6 Reserve chain

Definition 3.6.1

Suppose $(X_n, n = 0, 1, \ldots)$ is a discrete-time Markov chain. Define $Y_m = X_{n-m}$, $0 \leq m \leq n$. Y_m is the reverse chain.

Definition 3.6.2

If X has *one-step transition probabilities* P_{ij} and X has stationary distribution $\underline{\pi}$, then the one-step transition probabilities of Y are: $\tilde{P}_{ij} = \mathbb{P}(Y_{m+1} = j \mid Y_m = i) = \frac{\pi_j P_{ji}}{\pi_i}$.

Definition 3.6.3

If X satisfies local balance: $\pi_i P_{ij} = \pi_j P_{ji}$ and similarly for $\tilde{P}_{ij}$, then call X reversible in stationarity.

Theorem 3.6.4: Kolmogorov Cycle Condition

If X is *irreducible*, then X is reversible in stationarity *iff* $P_{ij} > 0 \Rightarrow P_{ji} > 0$ and for any "loop" $x_0, x_1, \ldots, x_n = x_0$ which has $\prod_{i=1}^n P_{x_i, x_{i+1}} > 0$, it holds that $\prod_{i=1}^n P_{x_i, x_{i-1}} = \prod_{i=1}^n P_{x_{i-1}, x_i}$.

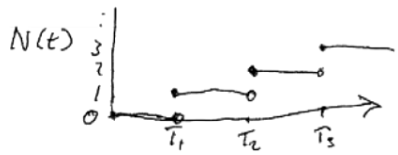
If Kolmogorov cycle condition holds for an irreducible Markov chain, then X has a stationary distribution.

Chapter 4

Poisson Processes

4.1 Counting processes

Suppose we have a random walk $(S_n, n \in \mathbb{N})$ with $S_0 = 0$ and $S_n = \sum_{k=1}^{n} B_k$, where $B_1, B_2, \ldots \overset{\text{iid}}{\sim} \text{Ber}(p)$. Such a walk is known as a *Bernoulli process*.



T_i : Arrival/Event times. A_i : Inter-arrival/event times.

Definition 4.1.1

A stochastic process $(N(t), t \geq 0)$ is called a *counting process* if:

(a) $N(0) = 0$

(b) $N(t) \in \mathbb{N}$ for all $t \geq 0$

(c) For $0 \leq s < t$, $N(t) - N(s)$ records the number of events in $(s, t]$

Theorem 4.1.2

For a counting process $(N(t), t \geq 0)$ with event times $T_1, T_2, \ldots$, for every $t \geq 0$, $n \in \mathbb{N}$, we have:

$$\{N(t) \geq n\} = \{T_n \leq t\}$$

4.2 Time-homogeneous Poisson process

Definition 4.2.1

A counting process $(N(t), t \geq 0)$ is called a *Poisson process with rate* λ if:

(a) For any interval $I \subseteq \mathbb{R}_+$, $N(I) \sim \text{Poisson}(\lambda|I|)$, where $|I|$ is the length of the interval.

(b) For every $m \in \{1, 2, \ldots\}$ and every selection of disjoint intervals $I_1, I_2, \ldots, I_m$, $N(I_1), N(I_2), \ldots, N(I_m)$ are independent.

Recall. If $X \sim \text{Poi}(\lambda)$ for fixed $\lambda > 0$, then $\mathbb{P}(X = x) = \frac{\lambda^x}{x!} e^{-\lambda}, \quad x \in \mathbb{N}$.

Example.

> Let $(N(t), t \geq 0)$ be a Poisson process with rate 2. How many events occur between time 0 & time 10? What is the probability that exactly 3 events occur in $[0, 10]$? What is $\mathbb{E}[N([0, 10])]$ and $\mathrm{Var}[N([0, 10])]$? What is the chance there are 3 events in $[0, 10]$ and 5 in $[18, 24]$?

$$N([0, 10]) \sim \mathrm{Poi}(20)$$
$$\mathbb{P}(N([0, 10]) = 3) = e^{-20}\frac{(20)^3}{3!}$$
$$\mathbb{E}[N([0, 10])] = \mathrm{Var}[N([0, 10])] = 20$$
$$N([18, 24]) \sim \mathrm{Poi}(2 \times (24 - 18)) = \mathrm{Poi}(12)$$
$$\mathbb{P}(N([0, 10]) = 3,\ N([18, 24]) = 5) = \mathbb{P}(N([0, 10]) = 3) \cdot \mathbb{P}(N([18, 24]) = 5) = e^{-20}\frac{20^3}{3!} \times e^{-12}\frac{12^5}{5!}$$

Example.

> What if the second interval were $[8, 14]$? What is the chance there are 3 events in $[0, 10]$ and 5 in $[8, 14]$?

To get the probability, just calculate each case and sum.

I_1	I_2	I_3
3	0	5
2	1	4
1	2	3
0	3	2

Recall. A random variable X with probability density function f, given by $f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$. The cdf of X is given by

$$F(x) = \mathbb{P}(X \leq x) = \int_0^x \lambda e^{-\lambda u}\, du = \left[-e^{-\lambda u}\right]_0^x = 1 - e^{-\lambda x}, \quad x \geq 0.$$

As consequence, the tail probability $\mathbb{P}(X > x)$ is exponentially decaying: $\mathbb{P}(X > x) = e^{-\lambda x}, \quad x > 0$. If $A_1, A_2, \ldots, A_n$ are n independent and identically distributed random variables following an Exponential(λ) distribution, then their sum $S_n = A_1 + A_2 + \cdots + A_n$ follows a Gamma(n, λ) distribution. The **gamma** distribution's density function is given by

$$f(x) = \frac{\lambda^\alpha x^{\alpha-1} e^{-\lambda x}}{\Gamma(\alpha)}, \quad x \geq 0,$$

where Γ is the Gamma function, defined as $\Gamma(\alpha) = \int_0^\infty u^{\alpha-1} e^{-u}\, du, \quad \alpha > 0$.

Lemma 4.2.2

Let $A_1, A_2, \ldots \sim$ iid Exp(λ). Then the corresponding counting process is a Poisson process with rate λ.

Proof.

$$\mathbb{P}(N(t) \geq n) = \mathbb{P}(T_n \leq t)$$
$$T_n \sim \mathrm{Gamma}(n, \lambda) \sim \mathrm{Erlang}(n, \lambda)$$
$$\mathbb{P}(T_n \leq t) = \int_0^t \frac{(\lambda s)^{n-1}}{(n-1)!}\lambda e^{-\lambda s}\, ds = 1 - \sum_{k=0}^{n-1}\frac{(\lambda t)^k}{k!}e^{-\lambda t} = \left[1 - \sum_{k=0}^{n-1}\mathbb{P}(X = k)\right], \quad X \sim \mathrm{Poi}(\lambda t)$$
$$= 1 - \mathbb{P}(X < n) = \mathbb{P}(X \geq n) \quad \therefore N(t) \sim \mathrm{Poi}(\lambda t)$$

□

For a Poisson process, $A_1, A_2, \ldots \sim$ iid Exp(λ)

Theorem 4.2.3

The exponential distribution has the **memoryless property**: if $X \sim \text{Exp}(\lambda)$, for $s, t \geq 0$, $\mathbb{P}(X > t + s \mid X > s) = \mathbb{P}(X > t)$.

Proof.

$$\mathbb{P}(X > t + s \mid X > s) = \frac{\mathbb{P}(X > t + s \cap X > s)}{\mathbb{P}(X > s)} = \frac{\mathbb{P}(X > t + s)}{\mathbb{P}(X > s)} = \frac{e^{-\lambda(t+s)}}{e^{-\lambda s}} = e^{-\lambda t} = \mathbb{P}(X > t)$$

□

Claim

Let $(N_t, t \geq 0)$ be a Poisson process with rate $\lambda > 0$. Then, for any fixed $s > 0$,

$$(N_t, 0 \leq t \leq s) \quad \text{and} \quad (N(t+s) - N(s), t \geq 0)$$

are independent Poisson processes with rate λ.

Proof for Claim.

Omitted

Claim

If

$$\mathbb{P}(N(t+h) = m \mid N(t) = n) = \begin{cases} 0, & m < n \\ 1 - \lambda h + o(h), & m = n \\ \lambda h + o(h), & m = n + 1 \\ o(h), & m > n + 1 \end{cases}$$

then $(N(t), t \geq 0)$ is a Poisson process with rate $\lambda > 0$. $g(h) \in o(h)$ if $\lim_{h \to 0} \frac{g(h)}{h} = 0$

Proof for Claim.

Denote $P_n(t) = \mathbb{P}(N(t) = n)$

$$P_n(t+h) = P_n(t) \cdot \mathbb{P}(N(t+h) = n \mid N(t) = n) + P_{n-1}(t) \cdot \mathbb{P}(N(t+h) = n \mid N(t) = n-1)$$

$$= P_n(t)(1 - \lambda h + o(h)) + P_{n-1}(t)(\lambda h + o(h))$$

$$\frac{P_n(t+h) - P_n(t)}{h} = -\lambda P_n(t) + \frac{o(h)}{h} P_n(t) + \lambda P_{n-1}(t) + \frac{o(h)}{h} P_{n-1}(t)$$

Taking $h \to 0$, for $t \geq 0$:

$$P_n'(t) = -\lambda P_n(t) + \lambda P_{n-1}(t)$$

We know that $P_0(0) = 1$, so $P_n(0) = 0$ for all $n \in \mathbb{N}_{\geq 1}$ and $P_{-m}(t) = 0$ for all $m \in \mathbb{N}_{\geq 1}$ and $t \geq 0$. Now consider the case of $n = 0$ in general:

$$P_0'(t) = -\lambda P_0(t) + \lambda \underbrace{P_{-1}(t)}_{=0} \quad \Rightarrow \quad P_0(t) = ce^{-\lambda t}, \quad \forall t \geq 0.$$

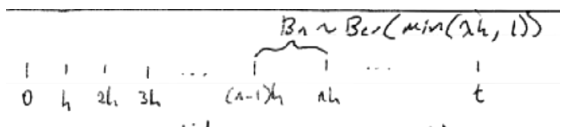
From $P_0(0) = 1$, we can see that $c = 1$. Hence, $P_0(t) = e^{-\lambda t}$, meaning

$$P_n(t) = e^{-\lambda t} \frac{(\lambda t)^n}{n!}, \quad \forall t \geq 0.$$

Example.

Example of $g(h)$:

$$e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \cdots = 1 - x + o(x)$$



Consider interval endpoints/times $0, h, \ldots, (n-1)h, nh, \ldots$ and to each pair a distance h apart, i.e. nh and $(n-1)h$, assign a random variable $B_n \sim \mathrm{Ber}(\min\{\lambda h, 1\})$. Then $B_1, B_2, \ldots \overset{\text{iid}}{\sim} \mathrm{Ber}(\min\{\lambda h, 1\})$ and $Y_n := B_1 + \cdots + B_n$ counts the number of events in $[0, nh]$; it has distribution

$$Y_n \sim \mathrm{Bin}(n, \min\{\lambda h, 1\}) \quad \text{with } n = \lfloor t/h \rfloor.$$

Claim

$N(t)$ and Y_n have approximately the same distribution, $\forall t \geq 0$

Proof for Claim.

$$\mathbb{P}(Y_n = m) = \binom{n}{m}(\lambda h)^m(1-\lambda h)^{n-m} \approx \binom{n}{m}\left(\frac{\lambda t}{n}\right)^m\left(1 - \frac{\lambda t}{n}\right)^{n-m} \quad \text{when } h \approx \frac{t}{n}$$

$$= \frac{n!}{(n-m)!m!}\left(\frac{\lambda t}{n}\right)^m\left(1 - \frac{\lambda t}{n}\right)^{n-m}$$

(3) As $n \to \infty$, $\left(1 - \frac{\lambda t}{n}\right)^{n-m} \to e^{-\lambda t}$

(2) $\lim_{n\to\infty}\left(\frac{\lambda t}{n}\right)^m = \left(\frac{\lambda t}{m!}\right)^m$

(1)

$$\lim_{n\to\infty}\frac{n!}{(n-m)!n^m} = \lim_{n\to\infty}\frac{n(n-1)\cdots(n-m+1)}{n^m} = \lim_{n\to\infty}\prod_{i=1}^{m-1}\left(1 - \frac{i}{n}\right) = 1$$

(1) Also, noting Stirling's formula:

$$\left(\frac{\lambda t}{m!}\right)^m e^{-\lambda t}$$

This suggests the Bernoulli Process (suitably scaled) converges "in distribution" to the Poisson Process.

Definition 4.2.4: Order Statistics

Let $U_1, U_2, \ldots, U_n \overset{\text{i.i.d.}}{\sim} \mathcal{U}[0, t]$ for some $t > 0$, $n \geq 1$. Let $U_{(1)} \leq \cdots \leq U_{(n)}$ be the corresponding *order statistic*

Let $T_1, \ldots, T_n$ be the event times of a Poisson process with rate $\lambda > 0$, and $(N(t), t \geq 0)$ be the corresponding counting process.

Claim

$(T_1, \ldots, T_n) \mid N(t) = n$ has the same distribution as $(U_{(1)}, \ldots, U_{(n)})$.

Proof for Claim.

Recall:

$$f_U(u) = \begin{cases} \frac{1}{t}, & u \in [0,t] \\ 0, & \text{otherwise} \end{cases} \qquad u \in [0,t]$$

$$\Rightarrow f_{U_1,\ldots,U_n}(u_1, \ldots, u_n) = \begin{cases} \left(\frac{1}{t}\right)^n, & (u_1, \ldots, u_n) \in [0,t]^n \\ 0, & \text{otherwise} \end{cases}$$

$$f_{U_{(1)},\ldots,U_{(n)}}(u_1, \ldots, u_n) = \begin{cases} \frac{n!}{t^n}, & 0 \le u_1 \le \cdots \le u_n \le t \\ 0, & \text{otherwise} \end{cases}$$

$$\mathbb{P}(A_{n+1} > t - t_n) = 1 - \mathbb{P}(A_{n+1} < t - t_n) = 1 - (1 - e^{-\lambda(t-t_n)}) = e^{-\lambda(t-t_n)}$$

$$f_{T_1,T_2,\ldots,T_n|N(t)=n}(t_1, \ldots, t_n) = \frac{e^{-\lambda(t-t_n)} \cdot \lambda e^{-\lambda(t_n - t_{n-1})} \cdots \lambda e^{-\lambda t_1}}{\frac{e^{-\lambda t}(\lambda t)^n}{n!}}$$

So:

$$= \frac{\lambda^n e^{-\lambda t}}{\frac{e^{-\lambda t}\lambda^n t^n}{n!}} = \begin{cases} \frac{n!}{t^n}, & 0 \le t_1 \le t_2 \le \cdots \le t_n < t \\ 0, & \text{otherwise} \end{cases}$$

Fact 4.2.5

Simulating a time-homogeneous Poisson Process with rate $\lambda > 0$ on interval $[0,t]$: Algorithm 2a:

- Set $T_0 = 0$, $n = 0$
- While $T_n < t$:
  - $T_{n+1} = T_n + \text{Exp}(\lambda)$
  - $n \leftarrow n + 1$
- Create $(N(s), s \in [0,t])$ from the arrival times

Algorithm 2b:

1. Generate $N \sim \text{Poisson}(\lambda t)$
2. Given $N = n$, generate $U_1, U_2, \ldots, U_n \overset{\text{i.i.d.}}{\sim} \mathcal{U}[0,t]$
3. Return $T_{(k)} = U_{(k)}$ for $1 \le k \le n$

Create $(N(s), s \in [0,t])$ from the arrival times.

4.3 Non-homogeneous Poisson process

Let $B_1, B_2, \ldots \overset{\text{iid}}{\sim} \text{Ber}(p_n)$ with $p_n = \lambda(nh)h$, where $\lambda : \mathbb{R}_{>0} \to \mathbb{R}_{>0}$. Doing the same limiting procedure as before, we arrive at another counting process known as a *non-homogeneous Poisson process*.

Definition 4.3.1: Non-homogeneous Poisson process

Given $\lambda : \mathbb{R}_+ \to \mathbb{R}_+$, we say the corresponding counting process $(N(t), t \geq 0)$ is a non-homogeneous Poisson process with rate function $\lambda(\cdot)$ if:

(a) $N([a,b]) \sim \text{Poi}\left(\int_a^b \lambda(s)\, ds\right)$

(b) # of points in disjoint intervals are independent.

Recall: For a time-homogeneous Poisson process with $\lambda > 0$:

(i) $\mathbb{P}(A_k > t) = e^{-\lambda t}, \quad t \geq 0$

(ii) For $i \neq j$, $f_{A_i,A_j}(s,t) = f_{A_i}(s) f_{A_j}(t) = \begin{cases} \lambda e^{-\lambda s} \lambda e^{-\lambda t}, & s,t \geq 0 \\ 0, & \text{otherwise} \end{cases}$

Example.

Let $A_1 = T_1$.

$$\mathbb{P}(T_1 > t) = \mathbb{P}(N(t) = 0) = e^{-\int_0^t \lambda(s) ds}$$

If $\lambda(s) = \lambda \Rightarrow \mathbb{P}(T_1 > t) = e^{-\lambda t}$

If $\lambda(s) = \lambda s \Rightarrow \mathbb{P}(T_1 > t) = e^{-\int_0^t \lambda s\, ds} = e^{-\frac{\lambda t^2}{2}}, \quad t \geq 0$

Inter-arrival times are no longer exponential!

Example.

Show that

$$f_{A_1,A_2}(s,t) = \begin{cases} \lambda(s)\lambda(t+s) e^{-\int_0^{t+s} \lambda(u) du}, & t,s \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

(In general) Interarrival times aren't independent.

Example.

Let $\lambda(s) = e^s$, $s \geq 0$ and define $I_1 := [0,1]$ and $I_2 = [5-\varepsilon, 5+\varepsilon]$, $1 > \varepsilon > 0$. We are interested in

$$\mathbb{P}(N(I_1) = 17, N(I_2) = 3).$$

Note that $I_1 \cap I_2 = \emptyset$. The relevant rates are

$$\lambda_1 = \int_0^1 e^s\, ds = e - 1$$

$$\lambda_2 = \int_{5-\varepsilon}^{5+\varepsilon} e^s\, ds = e^{5+\varepsilon} - e^{5-\varepsilon} = e^5(e^\varepsilon - e^{-\varepsilon}) = 2e^5 \sinh(\varepsilon).$$

Thus,

$$\mathbb{P}(N(I_1) = 17, N(I_2) = 3) = \mathbb{P}(N(I_1) = 17) \cdot \mathbb{P}(N(I_2) = 3) = e^{1-e} \frac{(e-1)^{17}}{17!} \cdot e^{-2e^5 \sinh(\varepsilon)} \frac{\left(2e^5 \sinh(\varepsilon)\right)^3}{3!}.$$

Definition 4.3.2: Superposition of Poisson Processes

Suppose $(N_1(t), t \geq 0)$ is a Poisson process with constant rate $\lambda_1 > 0$, and similarly $(N_2(t), t \geq 0)$ with rate $\lambda_2 > 0$, independent of N_1. $M(t) = N_1(t) + N_2(t)$ is a Poisson process with constant rate $\lambda_1 + \lambda_2$.

Theorem 4.3.3

Suppose $(N_1(t), t \geq 0)$ is a **non-homogeneous** Poisson process with rate function $\lambda_1 : \mathbb{R}_+ \to \mathbb{R}_+$, and $(N_2(t), t \geq 0)$ is a **non-homogeneous** Poisson process with rate function $\lambda_2 : \mathbb{R}_+ \to \mathbb{R}_+$. Then $M(t) = N_1(t) + N_2(t)$ is a **non-homogeneous Poisson process** with rate function $\lambda_1(\cdot) + \lambda_2(\cdot)$.

Definition 4.3.4: Thinning a Poisson Process

Let $(N(t), t \geq 0)$ be a Poisson process with rate function $\lambda : \mathbb{R}_{>0} \to \mathbb{R}_{>0}$ and parameter s (i.e. λ is a function of s). Suppose each event is kept with (time dependent) probability $p(s) \in (0, 1)$ and removed with probability $1 - p(s)$. The process composed of the accepted points is called the *thinned* process.

Theorem 4.3.5

Then the new, resultant processes (collections of accepted and rejected points), are Poisson processes with rates $p(s)\lambda(s)$ (accepted) and $(1 - p(s))\lambda(s)$ (rejected) and are independent.

Proof. Let $N(t)$ be the number of events of a Poisson process up to time t, with rate λ. Let $X_1(t)$ be the number of events that are **kept**. Let $X_2(t)$ be the number of events that are **rejected**. Let $Y_i \overset{\text{iid}}{\sim} \text{Ber}(p)$.

$$X_1(t) = Y_1 + Y_2 + \cdots + Y_{N(t)}$$

$$G_Y(z) = 1 - p + zp$$

$$G_{N(t)}(z) = e^{\lambda t(z-1)}$$

Now compute the generating function for $X_1(t)$:

$$G_{X_1(t)}(z) = \mathbb{E}_{N(t)}\left[\mathbb{E}_{Y|N(t)}\left[z^{\sum Y_i}\right]\right] = G_{N(t)}(G_Y(z))$$

$$= e^{\lambda t((1-p)+pz-1)} = e^{\lambda pt(z-1)}$$

Thus,

$$X_1(t) \sim \text{Poisson}(\lambda pt), \quad X_2(t) \sim \text{Poisson}(\lambda(1-p)t)$$

□

Suppose $(N(t), t \geq 0)$ is a constant rate Poisson process with parameter $\bar{\lambda} > 0$ satisfying

$$\bar{\lambda} \geq \sup_{t \geq 0} \lambda(t)$$

and suppose we thin the process with probability $p(t) = \lambda(t)/\bar{\lambda}$ so that $\lambda(t) = \bar{\lambda}p(t)$. We will be using this quantity in the following algorithm to simulate from Poisson processes.

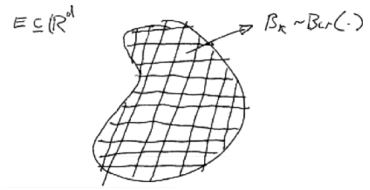
Fact 4.3.6

Algorithm 3 (Non-Homogeneous Poisson Process)
Input: $\lambda : \mathbb{R}_+ \to \mathbb{R}_+$, $\bar{\lambda}$ satisfying the condition above, and time interval $\mathcal{T} = [0, T]$.

1. Generate N on $[0, T]$ using either Algorithm 2a or 2b (constant-rate Poisson process with rate $\bar{\lambda}$).
2. Thin using $p(t) = \lambda(t)/\bar{\lambda}$.

4.4 Spatial Poisson Process (Poisson Random Measure)

Let $E \subseteq \mathbb{R}^d$, and $B_r \sim \text{Ber}(\cdot)$.



Definition 4.4.1: Poisson random measure

A process N is called a **Poisson random measure** with constant rate $\lambda > 0$ if:

(a) $N(\emptyset) = 0$

(b) For $A \subseteq E$, $N(A) \sim \text{Poi}(\lambda|A|)$

(c) For any disjoint sets $A_1, A_2, \ldots, A_n$, the random variables $N(A_1), N(A_2), \ldots, N(A_n)$ are independent

Definition 4.4.2: Non-homogeneous Poisson random measure

A random measure on $E \subset \mathbb{R}^d$ with rate function $\lambda : E \to \mathbb{R}_+$ is called a **non-homogeneous Poisson random measure (PRM)** if:

(a) $N(\emptyset) = 0$

(b) If $A \subseteq E$, then

$$N(A) \sim \text{Poisson}\left(\int_A \lambda(\underline{x})\, d\underline{x}\right)$$

(c) For disjoint sets $\mathcal{I}_1, \mathcal{I}_2, \ldots, \mathcal{I}_n \subset E$,

$$N(\mathcal{I}_1), N(\mathcal{I}_2), \ldots, N(\mathcal{I}_n) \text{ are independent.}$$

Fact 4.4.3

Algorithm 4a (Homogeneous PRM)
Ingredients: $E \subset \mathbb{R}^d$, $\lambda > 0$

(a) Generate $N \sim \text{Poisson}(\lambda|E|)$

(b) Given $N = n$, place points $\underline{x}_1, \ldots, \underline{x}_n$ uniformly in E

Fact 4.4.4

Algorithm 4b (Non-Homogeneous PRM)
Ingredients: $E \subset \mathbb{R}^d$, $\lambda : E \to \mathbb{R}^+$. Let $\bar{\lambda} > \lambda(\underline{x})$ for all $x \in E$, i.e., $\infty > \bar{\lambda} \geq \sup_{\underline{x} \in E} \lambda(\underline{x})$

(a) Simulate points from PRM with constant rate $\bar{\lambda}$ using Algorithm 4a.

(b) Thin each realised point $\underline{x}$ with probability

$$\rho(\underline{x}_i) = \frac{\lambda(\underline{x}_i)}{\bar{\lambda}}.$$

Example.

> Let N be a non-homogeneous PRM on $[0,1]^{10}$ with rate function
>
> $$\lambda(\underline{x}) = \prod_{i=1}^{10} e^{x_i},$$
>
> where $\underline{x} = (x_1, \ldots, x_{10}) \in [0,1]^{10}$.
>
> What is
>
> $$\mathbb{E}\left[N\left([0,\tfrac{1}{2}]^{10}\right)\right] \quad ?$$

Since $N\left([0,\frac{1}{2}]^{10}\right) \sim \text{Poisson}\left(\int_{[0,\frac{1}{2}]^{10}} \lambda(\underline{x})\, d\underline{x}\right)$, define:

$$\lambda^x := \int_{[0,\frac{1}{2}]^{10}} \prod_{i=1}^{10} e^{x_i}\, d\underline{x}.$$

We compute:

$$\lambda^x = \prod_{i=1}^{10} \left(\int_0^{1/2} e^{x_i}\, dx_i \right) = \prod_{i=1}^{10} \left[e^{x_i}\right]_{x_i=0}^{1/2} = \prod_{i=1}^{10} \left(\sqrt{e} - 1\right) = \left(\sqrt{e} - 1\right)^{10}.$$

Example.

> Tasks on a high-performance computing environment arrive at a rate of 1 task per second. Each task is successfully executed with probability 0.8, independently.
>
> (a) If 55 tasks are **successful** in 1 minute, what is the expected number of **unsuccessful** tasks in that minute?
>
> (b) If 55 tasks are **successful** in 1 minute, what is the expected number of tasks arriving in this time?

Let:

- $N(t)$: Poisson process for the number of tasks by time t
- $S(t)$: number of *successful* tasks by time t
- $F(t)$: number of *failed* tasks by time t

We have:

$$N(t) = S(t) + F(t),$$

and since each task is successful with probability 0.8, the failure rate is 0.2. So,

$$F(60) \sim \text{Poisson}(0.2 \cdot 60 \cdot 1) = \text{Poisson}(12).$$

Answer to (a):

$$\mathbb{E}[F(60) \mid S(60) = 55] = \mathbb{E}[F(60)] = 12.$$

Answer to (b):

$$\mathbb{E}[N(60) \mid S(60) = 55] = \mathbb{E}[F(60) \mid S(60) = 55] + 55 = 12 + 55 = 67.$$

Chapter 5

(Continuous Time) Markov Chains

5.1 Definition and Fundamental Results

Definition 5.1.1

Let the state space E be **countable** (usually $E \subseteq \mathbb{N}$ or $\mathbb{Z}$). We say a stochastic process is a *continuous-time Markov chain* if it satisfies the **Markov property**: if for any collection of times $0 \leq s_1 \leq s_2 \leq \cdots \leq s_n \leq s$ for $n \in \mathbb{N}$,

$$\mathbb{P}(X_{s+t} = j \mid X_s = i, X_{s_n} = i_n, \ldots, X_{s_0} = i_0) = \mathbb{P}(X_{s+t} = j \mid X_s = i)$$

for all $i_0, \ldots, i_n, i, j \in E$ and $s, t \geq 0$. If further

$$= \mathbb{P}(X_t = j \mid X_0 = i)$$

then we say the continuous-time Markov chain is **time-homogeneous**.

As for discrete time chains, we define the **transition probabilities** (from i to j) for the process to be

$$P_t(i,j) = \mathbb{P}(X_t = j \mid X_0 = i), \quad t > 0.$$

However, with continuous time, there is no "step" to speak of like with discrete time, so the transition probabilities must be specified separately for each particular time t, which we denote by the subscript on P above. For a **standard/honest** CTMC, this transition probability has the following property:

$$\lim_{t \to 0^+} P_t(i,j) = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

Theorem 5.1.2: Chapman–Kolmogorov Equations

$$P_{t+s}(i,j) = \sum_{k \in \Omega} P_s(i,k) P_t(k,j), \quad \forall s, t \geq 0,\ i, j \in E.$$

If we can collect the $P_t(i,j)$ into a matrix $P_t = \big(P_t(i,j),\ i,j \in E\big)$, then the Chapman-Kolmogorov equations can be expressed as

$$P_{t+s} = P_s P_t, \quad s, t \geq 0.$$

Furthermore, for a standard CTMC, Equation becomes $\lim_{t \to 0^+} P_t = I$.

Definition 5.1.3

Let W_t be the random amount of time we remain in a state given we are at time t:

$$\mathbb{P}(W_t > w \mid X_t = i) = \mathbb{P}(W_0 > w \mid X_0 = i) = h_i(w)$$

$$\begin{aligned}
h_i(u+v) &= \mathbb{P}_i(W_0 > u+v) \\
&= \mathbb{P}_i(W_0 > u, W_u > v) \\
&= \mathbb{P}_i(W_0 > u)\,\mathbb{P}_i(W_u > v \mid W_0 > u) \\
&= h_i(u)\,\mathbb{P}_i(W_u > v \mid W_0 > u, X_0 = i, X_u = i) \\
&= h_i(u)\,\mathbb{P}_i(W_u > v \mid X_u = i) \\
&= h_i(u)\,\mathbb{P}_i(W_0 > v \mid X_0 = i) \\
&= h_i(u)\,h_i(v), \quad u, v \geq 0
\end{aligned}$$

Claim

$h_i(u) = e^{-c_i u}$, $u > 0$ for some $c_i \geq 0$ (possibly $c_i = \infty$) is the **only** solution to the above equation that is a probability.

Proof for Claim.

Omitted

To completely specify the finite-dimensional distributions of an (honest) time-homogeneous CTMC, we need only specify 3 ingredients:

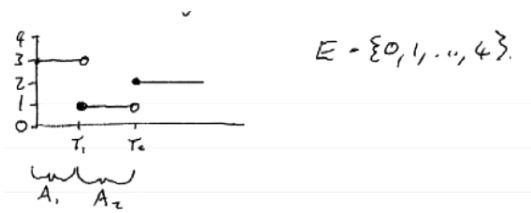
1. Initial distribution $\underline{\pi}^{(0)}$ over E.
2. Collection of exponential "holding times", q_i, $i \in E$.
3. The 1-step transition probabilities $(K(i,j))$, $i, j \in E$.

$K(i,j)$: given the chain "jumps" and is at state i, $K(i,j)$ is the probability it jumps to state j.

Fact 5.1.4

Algorithm 6
Ingredients: $T > 0$, $\underline{\pi}^{(0)}$, $\{q_i\}$, K

1. $t = 0, \quad X_0 \sim \underline{\pi}^{(0)}$
2. While $t \leq T$ do:
   - $y \sim \text{Exp}(q_{X_t})$
   - $X_{t+y} \sim K(X_t, \cdot)$
   - $t \leftarrow t + y$



- T_i: jump times
- A_i: holding times / sojourn times

- $A_i \sim \text{Exp}(q_{X_{T_{i-1}}})$
- If $q_x = +\infty$, we call the state x *instantaneous*.
- K_{xy}: one-step transition probabilities conditioned on $K_{xx} = 0$
- $K = (K_{xy})_{x,y \in E}$ is the one-step transition probability matrix of the "jump chain".

Recall. the expectation of $\text{Exp}(\lambda)$ is $\frac{1}{\lambda}$

Example.

> $E = \mathbb{N}$ is Time-homogeneous DTMC with $K_{x,x+1} = 1 \quad \forall x \in \mathbb{N}, \quad q_x = \lambda$.

Example.

> Let $E = \mathbb{N}$, $K_{x,x+1} = 1$ and $q_x = 2^x$. Define T_n to be the time of the nth jump.

Then

$$\mathbb{E}_0 T_n = \mathbb{E}_0 A_1 + \cdots + \mathbb{E}_{n-1} A_n = 1 + \frac{1}{2} + \cdots + \frac{1}{2^{n-1}} = \sum_{x=0}^{n-1} \frac{1}{2^x} = \frac{1-(1/2)^n}{1-1/2} = 2 - 2^{1-n}, \quad n \in \mathbb{N}_{\geq 1}.$$

Hence, $\lim_{n\to\infty} \mathbb{E}_0 T_n = 2$, meaning

$$\mathbb{E}_0 \lim_{n\to\infty} T_n = \lim_{n\to\infty} \mathbb{E}_0 T_n = 2 \quad \Rightarrow \quad \mathbb{P}\left(\lim_{n\to\infty} T_n = \infty\right) = 0,$$

where the first equality follows from Equation, the defining property of a time homogeneous CTMC.

The above situation can be described as an "explosion" in the process. We would "like" the process to be **regular**:

$$\mathbb{P}\left(\lim_{n\to\infty} T_n = \infty\right) = 1.$$

Definition 5.1.5

We say an **honest, regular CTMC with no instantaneous states** satisfies the "standard conditions".

Example.

> Consider queueing system where people arrive in the queue according to a Poisson process with rate $\lambda > 0$ and there is a single server whose service times are exponentially distributed: $\text{Exp}(\mu)$. Such a system is known as an **M/M/1** system (Markovian arrival process/Markovian service process/1 server).
>
>
>
>
> We want to compute:
>
> $$q_x = ? \qquad K_{x,x+1} = ? \qquad K_{x,x-1} = ? \quad \text{for } x = 1, 2, \ldots$$

Let $(X_t) \in \mathbb{Z}_{\geq 0}$, where X_t is the number of people in the queue, including the one being served. Our starting quantities are $E = \mathbb{N}$, $q_0 = \lambda$ and $K_{0,1} = 1$. We know from above that $A \sim \text{Exp}(\lambda)$ and $S \sim \text{Exp}(\mu)$ (I'm assuming these stand for "arrival times" and "service times"), with the two of them independent. In order to find q_x and the various κ quantities, we now wish to know if $\min\{A, S\} \sim \text{Exp}(\lambda+\mu)$. We have

$$\mathbb{P}(\min\{A,S\} > t) = \mathbb{P}(A > t, S > t) = \mathbb{P}(A > t)\mathbb{P}(S > t) = e^{-\lambda t} e^{-\mu t} = e^{-(\lambda+\mu)t},$$

so indeed, $\min\{A, S\} \sim \text{Exp}(\lambda + \mu)$. Hence,

$$\mathbb{P}(A < S) = k_{x,x+1} = \frac{\lambda}{\lambda + \mu}, \quad \mathbb{P}(S < A) = k_{x,x-1} = \frac{\mu}{\lambda + \mu}.$$

Finally, this implies that $q_x = \lambda + \mu$.

5.2 Q Matrix

Let $(X_t, t \geq 0)$ be a CTMC with state space E, event times $\{T_i\}_{i \geq 1}$, holding times $\{q_x\}$ and jump chain one step transition matrix K. Recall from the previous section that $P_t(x, y) = \mathbb{P}(X_t = y \mid X_0 = x) = \mathbb{P}_x(X_t = y)$. Noting that $(T_1 \mid X_0 = x) \sim \text{Exp}(q_x)$, we have

$$\begin{aligned}
P_h(x, y) &= \mathbb{P}(X_h = y \mid X_0 = x) \\
&= \int_0^\infty q_x e^{-q_x u} \mathbb{P}(X_h = y \mid X_0 = x, T_1 = u)\, du \\
&= \int_0^h q_x e^{-q_x u} \mathbb{P}(X_h = y \mid X_0 = x, T_1 = u)\, du \\
&\quad + \int_h^\infty q_x e^{-q_x u} \mathbb{P}(X_h = y \mid X_0 = x, T_1 = u)\, du \\
&= \begin{cases} \int_0^h q_x e^{-q_x u} \mathbb{P}(X_h = y \mid X_0 = x, T_1 = u)\, du, & x \neq y \\ \int_0^h q_x e^{-q_x u} \mathbb{P}(X_h = y \mid X_0 = x, T_1 = u)\, du + \int_h^\infty q_x e^{-q_x u}\, du, & x = y \end{cases} \\
&= \begin{cases} \int_0^h q_x e^{-q_x u} \sum_{z \in E} K_{x,z} P_{h-u}(z, y)\, du, & x \neq y \\ \int_0^h q_x e^{-q_x u} \sum_{z \in E} K_{x,z} P_{h-u}(z, y)\, du + e^{-q_x h}, & x = y \end{cases}
\end{aligned}$$

Approximating for small h:

$$\approx \begin{cases} h \cdot q_x e^{-q_x h} \sum_{z \in E} K_{x,z} P_0(z, y), & y \neq x \\ h \cdot q_x e^{-q_x h} \sum_{z \in E} K_{x,z} P_0(z, y) + e^{-q_x h}, & y = x \end{cases}$$

Since

$$e^x = 1 - x + \frac{x^2}{2} - \cdots = 1 - x + o(h)$$

We get

$$= \begin{cases} h q_x (1 - q_x h + o(h)) \sum_{z \in E} K_{x,z} \mathbb{I}(z = y), & y \neq x \\ h q_x (1 - q_x h + o(h)) \sum_{z \in E} K_{x,z} \mathbb{I}(z = y) + (1 - q_x h + o(h)), & y = x \end{cases}$$

$$= \begin{cases} h q_x K_{xy} + o(h), & y \neq x \\ h q_x K_{xy} + 1 - q_x h + o(h), & y = x \end{cases}$$

$$= \begin{cases} h q_x K_{xy} + o(h), & x \neq y \\ 1 - q_x h + o(h), & x = y \end{cases}$$

Now taking the derivative:

$$\begin{aligned}
P_0'(x, y) &= \lim_{h \to 0} \frac{P_h(x, y) - P_0(x, y)}{h} \\
&= \lim_{h \to 0} \frac{(1 - \mathbb{I}(x = y)) h q_x K_{xy} + \mathbb{I}(x = y)(1 - q_x h) - \mathbb{I}(x = y)}{h} \\
&= \mathbb{I}(x \neq y) q_x K_{xy} + \mathbb{I}(x = y)(-q_x) \\
P_0'(x, y) &= \mathbb{I}(x \neq y)\, q_x\, K_{xy} + \mathbb{I}(x = y)(-q_x)
\end{aligned}$$

P_0' is the "rate of probability flux at any time". Defining $q_{x,y} := q_x \kappa_{x,y}$, we have

$$\sum_{y \neq x} q_{x,y} = \sum_{y \neq x} q_x K_{x,y} = q_x \sum_{y \neq x} K_{x,y} = q_x.$$

Now define $Q := (q_{x,y},\ x, y \in E)$ (called the **generator matrix**) and $q_{x,x} := -q_x$ (the off-diagonal elements of Q are the rates of going from state i to state j, while the diagonal elements $q_{x,x}$ are the rates of leaving the given state x). Then

$$\sum_{y \in E} q_{x,y} = q_{x,x} + \sum_{y \neq x} q_{x,y} = -q_x + q_x = 0.$$

Q, if we can view it as a matrix, is *singular*. Hence Q is not invertible and has 0 as an eigenvalue.

Furthermore, we can write

$$K_{x,y} = \frac{q_{xy}}{q_x} + \mathbf{1}_{\{x=y\}} = \begin{cases} \mathbb{P}(Y_{n+1} = y \mid Y_n = x), & x \neq y, \\ \mathbb{P}(Y_{n+1} = x \mid Y_n = x), & x = y \end{cases} = \begin{cases} q_{xy}/q_x, & x \neq y, \\ 0, & x = y \text{ and } x \text{ is not absorbing}, \\ 1, & x = y \text{ and } x \text{ is absorbing}. \end{cases}$$

From the above, it follows that the model primitives (i.e., the information needed to specify the stochastic process) for a time homogeneous CTMC are $\pi^{(0)}$ and Q.

Theorem 5.2.1: Kolmogorov Backward Equation

P_t satisfies $P_t' = QP_t$ for all $t > 0$.

Theorem 5.2.2: Kolmogorov Forward Equation

Under additional conditions (like finite $|\Omega|$ or the "standard conditions"), then $P_t' = P_t Q$ for all $t > 0$.

Theorem 5.2.3

Under additional conditions, the solution to both KBE and KFE is unique and can be written as

$$P_t = e^{tQ} \quad \text{for all } t > 0.$$

Proof.

$$A : n \times n \text{ matrix exponential} \quad e^A = \sum_{k=0}^{\infty} \frac{A^k}{k!}$$

$$P_t = e^{tQ} = \frac{I}{0!} + \frac{tQ}{1!} + \frac{t^2 Q^2}{2!} + \cdots$$

$$P_t' = Q + tQ^2 + \frac{t^2}{2!} Q^3 + \cdots$$

$$= Q\left[I + tQ + \frac{t^2}{2!} Q^2 + \cdots\right] = QP_t$$

$$= \left[I + tQ + \frac{t^2}{2} Q + \cdots\right] Q = P_t Q$$

□

5.3 Limiting Distributions

Let $\underline{\pi}^{(0)}$ be the initial distribution of the process. Then the distribution at time t is $\underline{\pi}^{(t)} = \underline{\pi}^{(0)} P_t$. The **limiting distribution** is then

$$\underline{\pi}^{(\infty)} = \lim_{t \to \infty} \underline{\pi}^{(t)} = \lim_{t \to \infty} \underline{\pi}^{(0)} P_t \text{ "=" } \underline{\pi}^{(0)} \lim_{t \to \infty} P_t.$$

The "=" indicates that for an infinite state space, there are some additional conditions required for the equality to hold (but it still does).

Definition 5.3.1: stationary distribution

If a distribution $\underline{\pi}$ satisfies $\underline{\pi} P_t = \underline{\pi}$ for all $t \geq 0$ and $\underline{\pi}\mathbf{1} = 1$, it is called a **stationary** distribution.

Spectral Decomposition of Q

Suppose $|E| = p < \infty$. Let $\lambda_1, \ldots, \lambda_p$ be the eigenvalues and $\underline{v}_1, \underline{v}_2, \ldots, \underline{v}_p$ and $\underline{w}_1, \underline{w}_2, \ldots, \underline{w}_p$ be the corresponding right and left eigenvectors, where they are normalized so that $\underline{w}_i \cdot \underline{v}_i = 1$. Let $R_i = \underline{v}_i \underline{w}_i$.

We can write:

$$Q = \lambda_1 R_1 + \lambda_2 R_2 + \cdots + \lambda_p R_p$$

$$Q^2 = \lambda_1^2 R_1 + \lambda_2^2 R_2 + \cdots + \lambda_p^2 R_p$$

$$P_t = e^{tQ} = \frac{I}{0!} + \frac{tQ}{1!} + \frac{t^2 Q^2}{2!} + \cdots$$

$$= e^{\lambda_1 t} R_1 + \cdots + e^{\lambda_p t} R_p$$

$$\lambda_k = -\alpha_k + i\beta_k, \quad k = 2, \ldots, p$$

$$\lambda_1 = 0$$

$$P_t = R_1 + \sum_{k=2}^{\infty} e^{-\alpha_k t} \left(\cos(\beta_k t) + i \sin(\beta_k t)\right) R_k$$

For a discrete state space, we conclude

$$\lim_{t \to \infty} P_t = R_1$$

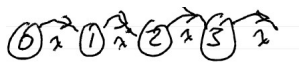
$$\therefore \pi^{(\infty)} = \lim_{t \to \infty} \pi^{(t)}$$

$$= \lim_{t \to \infty} \pi^{(0)} P_t$$

$$= \pi^{(0)} \left(\lim_{t \to \infty} P_t \right) = \pi^{(0)} R_1$$

$$= \text{any row of } R_1$$

Example.

> Poisson process
>
> 
>

$$Q = \begin{bmatrix} -\lambda & \lambda & 0 & 0 & \cdots & 0 \\ 0 & -\lambda & \lambda & 0 & \cdots & 0 \\ 0 & 0 & -\lambda & \lambda & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \ddots & \vdots \end{bmatrix}$$

The weights are not probabilities.

Example.

For the M/M/1 Queue with arrival rate λ and service rate μ, state $\Omega = \mathbb{N}$ and the rate transition graph.

$$Q = \begin{pmatrix} -\lambda & \lambda & 0 & 0 & \cdots \\ \mu & -(\lambda+\mu) & \lambda & 0 & \cdots \\ 0 & \mu & -(\lambda+\mu) & \lambda & \cdots \\ \vdots & \ddots & \ddots & \ddots & \ddots \end{pmatrix}.$$

$$\begin{tikzcd} 0 \arrow[r, bend left, "\lambda"] & 1 \arrow[l, bend left, "\mu"] \arrow[r, bend left, "\lambda"] & 2 \arrow[l, bend left, "\mu"] \arrow[r, bend left, "\lambda"] & \cdots \arrow[l, bend left, "\mu"] \end{tikzcd}$$

Example.

Consider a two state CTMC with transition rate graph as shown below (let $\lambda, \mu > 0$).

$$\begin{tikzcd} 0 \arrow[r, bend left, "\lambda"] & 1 \arrow[l, bend left, "\mu"] \end{tikzcd}$$

This process has corresponding Q-matrix

$$Q = \begin{pmatrix} -\lambda & \lambda \\ \mu & -\mu \end{pmatrix} \Rightarrow \operatorname{tr}(Q) = \lambda_1 + \lambda_2 = \lambda_2 = -(\lambda+\mu),$$

where we have taken $\lambda_1 = 0$ to be our zero eigenvalue. Then

$$P_t = R_1 + e^{-(\lambda+\mu)t} R_2 \Rightarrow P_0 = I = R_1 + R_2.$$

Furthermore, $Q = P_0' = -(\lambda+\mu)R_2$, so

$$R_2 = \frac{1}{\lambda+\mu}\begin{pmatrix} \lambda & -\lambda \\ -\mu & \mu \end{pmatrix} \Rightarrow R_1 = I - R_2 = \frac{1}{\lambda+\mu}\begin{pmatrix} \mu & \lambda \\ \mu & \lambda \end{pmatrix}.$$

Therefore, $\lim_{t\to\infty} P_t = R_1$ and the limiting distribution is

$$\underline{\pi}^{(\infty)} = \underline{\pi}^{(0)} R_1 = \frac{1}{\lambda+\mu}\begin{pmatrix} \mu & \lambda \end{pmatrix}.$$

Theorem 5.3.2: Stationary Distribution

Under certain conditions (such as finite state space cardinality or an irreducible positive recurrent chain), if $\underline{\pi}$ is a distribution on the state space Ω and $\underline{\pi}Q = 0$, then $\underline{\pi}P_t = \underline{\pi}$ for all $t \geq 0$.

Proof. If $P_t = e^{tQ} = \sum_{n\geq 0}(tQ)^n/n!$, then

$$\underline{\pi} = \underline{\pi} + \underline{\pi}Q\sum_{n=1}^{\infty}\frac{t^n Q^{n-1}}{n!} = \underline{\pi}\left(I + \sum_{n=1}^{\infty}\frac{(tQ)^n}{n!}\right) = \underline{\pi}\sum_{n=0}^{\infty}\frac{(tQ)^n}{n!} = \underline{\pi}P_t.$$

□

5.4 Classification of States for CTMCs

Let T be the time of the first exit from the current state and define

$$\tau_y := \inf\{t > T : X_t = y\}, \quad \tau_y^{(k)} := \inf\left\{t > \tau_y^{(k-1)} : X_t = y\right\}.$$

Also, let $\gamma_{xy} := \mathbb{P}_x(\tau_y < \infty)$. As with DTMCs, we say $(x, y \in \Omega)$

- $x \to y$ if $\gamma_{xy} > 0$.
- $x \leftrightarrow y$ if $x \to y$ and $y \to x$.
- x is transient if $\mathbb{P}_x(\tau_x < \infty) = \gamma_{xx} < 1$.
- x is recurrent if $\gamma_{xx} = 1$.
  - Positive recurrent if $\mathbb{E}_x T_x < \infty$.
  - Negative recurrent if $\mathbb{E}_x T_x = \infty$.
- No states are periodic or aperiodic in a CTMC.
- The notions of communicating class, irreducible, closed and absorbing are the same as with DTMCs.

Example.

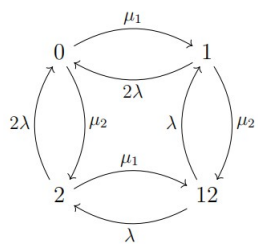
(Repairmen Problem). Suppose there are two machines failing at different exponential rates (machines 1 and 2, failing at exponential rates μ_1 and μ_2) and that there are two repairmen, repairing at the same exponential rate λ. If only one machine has failed, both men will work on it. If both machines have failed, each man will work on one separate machine. Initially, all machines are working.

We name the states of this system as follows:

- 0: both machines working.
- 1: machine 1 failed.
- 2: machine 2 failed.
- 12: both machines failed.

It follows that the state space of our system is $\Omega = \{0, 1, 2, 12\}$ and it has initial distribution $\pi^{(0)} = (1, 0, 0, 0)$. The generator matrix is

$$Q = \begin{pmatrix} -(\mu_1 + \mu_2) & \mu_1 & \mu_2 & 0 \\ 2\lambda & -(2\lambda + \mu_2) & 0 & \mu_2 \\ 2\lambda & 0 & -(2\lambda + \mu_1) & \mu_1 \\ 0 & \lambda & \lambda & -2\lambda \end{pmatrix}.$$



We are now interested in the stationary distribution for this process. This process is irreducible, recurrent and has finite state space, so Theorem applies. Hence, we solve $\boldsymbol{\pi Q = 0}$ and $\boldsymbol{\pi 1 = 1}$:

$$\begin{cases} -\pi_0(\mu_1 + \mu_2) + 2\lambda\pi_1 + 2\lambda\pi_2 = 0 \\ \pi_0\mu_1 - \pi_1(2\lambda + \mu_2) + \lambda\pi_{12} = 0 \\ \pi_0\mu_2 - \pi_2(2\lambda + \mu_1) + \lambda\pi_{12} = 0 \\ \pi_0 + \pi_1 + \pi_2 + \pi_{12} = 1 \end{cases}$$

$$\Rightarrow \underline{\pi} = \frac{1}{\lambda^2 + \mu_1\lambda + \mu_2\lambda + \mu_1\mu_2} \begin{pmatrix} \lambda^2 & \mu_1\lambda & \mu_2\lambda & \mu_1\mu_2 \end{pmatrix}.$$

Theorem 5.4.1

Let X be an irreducible, recurrent CTMC with state space Ω. Then

$$\lim_{t\to\infty} \mathbb{P}_x(X_t = y) = \pi_y, \qquad \forall x \in \Omega,$$

for some collection of numbers $\boldsymbol{\pi} := (\pi_y, y \in \Omega)$ such that $\pi_y \geq 0$ with strict inequality if y is positive recurrent. In the latter case, $\boldsymbol{\pi}$ solves $\boldsymbol{\pi Q} = 0$ and $\boldsymbol{\pi 1} = 1$ and thus is a stationary distribution.

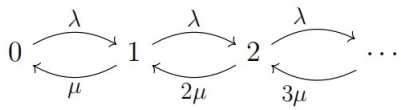
$Q = D(K - I)$, where I is the usual identity matrix, K is the jump chain one step transition matrix and $D = \operatorname{diag}(q_x)_{x\in\Omega}$. Let $\boldsymbol{\pi}$ solve $\boldsymbol{\pi Q} = 0$ and $\boldsymbol{v}$ solve $\boldsymbol{v}K = \boldsymbol{v}$. Then $\pi_x \propto v_x/q_x$ for all $x \in \Omega$. If this expression is normalisable, then

$$\pi_x = \frac{v_x/q_x}{\sum_{y\in\Omega} v_y/q_y}, \qquad \forall x \in \Omega.$$

It is not normalisable if $\sum_{y\in\Omega} v_y/q_y = \infty$.

Example.

> (M/M/∞ Queue). A queueing system where customers arrive according to a Poisson process with rate λ and are served by an infinite number of servers, all with service times that are distributed according to Exp(μ).



As with DTMCs, there are global and local balance conditions for a CTMC stationary distribution. The **global balance** equations are, from $\boldsymbol{\pi Q} = \mathbf{0}$,

$$\sum_{y\in\Omega} \pi_y q_{yx} = 0, \qquad \forall y \in \Omega.$$

The **local/detailed balance** equations are

$$\pi_y q_{yx} = \pi_x q_{xy}, \qquad \forall x, y \in \Omega.$$

Also as with DTMCs, local balance implies global balance:

$$\sum_{y\in\Omega} \pi_y q_{yx} = \sum_{y\in\Omega} \pi_x q_{xy} = \pi_x \sum_{y\in\Omega} q_{xy} = 0.$$

Theorem 5.4.2: Kolmogorov Cycle Condition

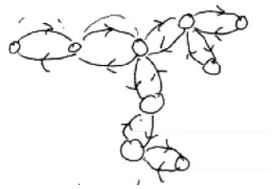
*For any $n \in \mathbb{N}$ and any $x_0, x_1, \ldots, x_n, x_0$ for which $q(x_i, x_{i+1}) > 0$ for $i \in \{0, \ldots, n-1\}$, $q(x_n, x_0) > 0$, $q(x_{i+1}, x_i) > 0$ for $i \in \{0, \ldots, n-1\}$ and $q(x_0, x_n)$ (this is a "cycle"), the **Kolmogorov Cycle Condition (KCC)** holds if*

$$\prod_{i=1}^{n} q(x_{i-1}, x_i) q(x_n, x_0) = \prod_{i=1}^{n} q(x_i, x_{i-1}) q(x_0, x_n).$$

Theorem 5.4.3

A distribution $\boldsymbol{\pi} = (\pi_x, x \in \Omega)$ satisfies the local balance equations for Q if and only if the Kolmogorov Cycle Condition holds.

In particular, tree-like transition graphs satisfy the KCC (i.e. transition graphs that would be trees if there was only one edge between each node).



5.5 Birth-Death Processes

$$\begin{tikzcd} 0 \arrow[r, bend left, "\lambda_0"] & 1 \arrow[l, bend left, "\mu_1"] \arrow[r, bend left, "\lambda_1"] & 2 \arrow[l, bend left, "\mu_2"] \arrow[r, bend left, "\lambda_2"] & \cdots \arrow[l, bend left, "\mu_3"] \end{tikzcd}$$

Let the state space be $\Omega = \mathbb{N}$, $\{\lambda_i\}_{i\in\mathbb{N}}$ be the birth rates and $\{\mu_i\}_{i=1}^{\infty}$ be the death rates, where $\lambda_i, \mu_i > 0$ for all $i \in \mathbb{N}$. Since the transition rate graph is tree-like, we can solve the local balance equations to find a stationary distribution:

$$\begin{aligned} \pi_0\lambda_0 = \pi_1\mu_1 \quad &\Longrightarrow \quad \pi_1 = \frac{\lambda_0}{\mu_1}\pi_0 \\ \pi_1\lambda_1 = \pi_2\mu_2 \quad &\Longrightarrow \quad \pi_2 = \frac{\lambda_1}{\mu_2}\pi_1 = \frac{\lambda_1\lambda_0}{\mu_2\mu_1}\pi_0 \\ &\vdots \end{aligned}$$

Continuing in this manner, we see that

$$\pi_n = \pi_0 \cdot \prod_{k=1}^{n} \frac{\lambda_{k-1}}{\mu_k}, \quad n \in \mathbb{N}.$$

We want to normalise:

$$\sum_{n=0}^{\infty} \pi_n = 1 \Longleftrightarrow \pi_0 \left(1 + \sum_{n=1}^{\infty}\prod_{k=1}^{n} \frac{\lambda_{k-1}}{\mu_k}\right) = 1 \Longrightarrow \pi_0 = \left(1 + \sum_{n=1}^{\infty}\prod_{k=1}^{n} \frac{\lambda_{k-1}}{\mu_k}\right)^{-1}.$$

Define

$$S := \sum_{n=1}^{\infty}\prod_{k=1}^{n} \frac{\lambda_{k-1}}{\mu_k}.$$

Then if S diverges, we cannot normalise and there is no stationary distribution. If S converges, we can and then the stationary distribution is given by

$$\pi_n = \left(1 + \sum_{n=1}^{\infty}\prod_{k=1}^{n} \frac{\lambda_{k-1}}{\mu_k}\right)^{-1} \prod_{k=1}^{n} \frac{\lambda_{k-1}}{\mu_k}.$$

Example.

> (M/M/1 Queue). Consider an M/M/1 queue where customers arrive according to a Poisson process with rate λ and are served by a single server, with service times that are distributed according to Exp(μ).
>
> $$\begin{tikzcd} 0 \arrow[r, bend left, "\lambda"] & 1 \arrow[l, bend left, "\mu"] \arrow[r, bend left, "\lambda"] & 2 \arrow[l, bend left, "\mu"] \arrow[r, bend left, "\lambda"] & \cdots \arrow[l, bend left, "\mu"] \end{tikzcd}$$

Define $\varrho := \lambda/\mu$. Then

$$S = \sum_{n=1}^{\infty} \varrho^n = \begin{cases} \frac{1}{1-\varrho} - 1, & \varrho < 1 \iff \lambda < \mu, \\ \infty, & \varrho \geq 1. \end{cases}$$

Hence,

$$\pi_n = \frac{\varrho^n}{\sum_{n=0}^{\infty} \varrho^n} = \frac{\varrho^n}{\frac{1}{1-\varrho}} = (1-\varrho)\varrho^n, \quad n \in \mathbb{N}.$$

This is a geometric stationary distribution.

Example.

> (M/M/∞ Queue). Consider an M/M/∞ queue where customers arrive according to a Poisson process with rate λ and are served by an infinite number of servers, all with service times that are distributed according to Exp(μ).
>
> $$\begin{tikzcd} 0 \arrow[r, bend left, "\lambda"] & 1 \arrow[l, bend left, "\mu"] \arrow[r, bend left, "\lambda"] & 2 \arrow[l, bend left, "2\mu"] \arrow[r, bend left, "\lambda"] & \cdots \arrow[l, bend left, "3\mu"] \end{tikzcd}$$

Recall. The Taylor (Maclaurin) series expansion for e^x around $x = 0$ is:

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

Define $\varrho := \lambda/\mu$. Then

$$1 + S = \sum_{n=0}^{\infty} \prod_{k=1}^{n} \frac{\lambda_{k-1}}{\mu_k} = \sum_{n=0}^{\infty} \frac{\lambda \cdots \lambda}{\mu \cdot 2\mu \cdot 3\mu \cdots n\mu} = \sum_{n=0}^{\infty} \frac{\varrho^n}{n!} = e^{\varrho} < \infty,$$

regardless of λ and μ. Note that in this example, $\mu_k = k\mu$, as can be seen from the transition rate graph. Hence,

$$\pi_n = \frac{\varrho^n}{n! e^{\varrho}} = e^{-\varrho} \frac{\varrho^n}{n!}, \quad n \in \mathbb{N}.$$

which is a Poisson distribution.

Example.

> At a hospital, patients arrive at a rate of $\lambda = 4$ per hour. There are four beds that can be allocated at any time. If new patients arrive when the beds are full, they are referred to another wing of the hospital.
>
> When there are 2 or fewer patients, a single doctor assesses them, and the service time is on average 30 minutes ($\mu = 2$ per hour).
>
> If there are 3 or more patients, another doctor assists, reducing the service time to 15 minutes ($\mu = 4$ per hour).
>
> What proportion of time are both doctors idle?

Define the state space and parameters:

$$\lambda_n = \begin{cases} 4, & n = 1, 2, 3, 4 \\ 0, & \text{otherwise} \end{cases} \qquad \mu_n = \begin{cases} 2, & n = 1, 2 \\ 4, & n = 3, 4 \\ 0, & \text{otherwise} \end{cases}$$

From the balance equations:

$$\lambda_0\pi_0 = \mu_1\pi_1 \Rightarrow 4\pi_0 = 2\pi_1 \Rightarrow \pi_1 = 2\pi_0$$

$$\pi_2 = \frac{4}{2}\pi_1 = 2\pi_1 = 4\pi_0$$

$$\pi_3 = \frac{4}{4}\pi_2 = \pi_2 = 4\pi_0$$

$$\pi_4 = \frac{4}{4}\pi_3 = \pi_3 = 4\pi_0$$

Normalization:

$$\pi_0(1+2+4+4+4) = 1 \Rightarrow \pi_0 = \frac{1}{15}$$

Proportion of time both doctors are idle: this happens when the system is in states 0, 1, or 2:

$$\pi_0 + \pi_1 + \pi_2 = \frac{1}{15} + \frac{2}{15} + \frac{4}{15} = \frac{7}{15}$$

Chapter 6

Probabilistic Measure Theory

6.1 Preliminary

Definition 6.1.1

Ω: abstract set

Example.

$\mathbb{N},\ \mathbb{Q},\ \varnothing$: no, $\mathbb{Z},\ \mathbb{R},\ \mathbb{R}_+,\ \mathbb{R}^n,\ \cdots,\ \{\text{Sunny}, \text{Cloudy}, \text{Rainy}\}$.

$$\begin{aligned} A &= \{\omega\}, && \text{where } \omega \in \Omega. \ \text{ singleton} \\ B &= \{\omega_1, \ldots, \omega_{10}\} \\ C &= \{\omega \in \Omega : \omega \leq 1\}. \end{aligned}$$

Definition 6.1.2: Set operation

Given $A, B, A_1, A_2, \ldots,$ then

$$\begin{aligned} A \cup B &\quad \text{union} \\ A \cap B &\quad \text{intersection} \\ A^c &\quad \text{complement} \\ \varnothing &\quad \text{empty set} \end{aligned}$$

$$A \setminus B \equiv A \cap B^c \quad \text{set difference}$$

$$A \triangle B \quad \text{symmetric diff.}$$

$$\bigcup_{n=1}^{\infty} A_n, \qquad \bigcap_{n=1}^{\infty} A_n$$

$$A \subseteq B \qquad \text{A is a subset of } B$$

$$A \subset B \qquad \text{A is a strict subset of } B$$

Definition 6.1.3

If $A \cap B = \varnothing$ we say A and B are disjoint. A sequence of subsets of Ω, $A_1, A_2, \ldots,$ are called mutually disjoint if $A_i \cap A_j = \varnothing$, for any $i \neq j$.

Theorem 6.1.4: De Morgan's Laws

$$\left(\bigcup_{n\geq 1} A_n\right)^c = \bigcap_{n\geq 1} A_n^c$$
$$\left(\bigcap_{n\geq 1} A_n\right)^c = \bigcup_{n\geq 1} A_n^c$$

Definition 6.1.5

If $A_1, A_2, \ldots$ is a sequence of mutually disjoint subsets of Ω s.t.

$$\bigcup_{n=1}^{\infty} A_n = \Omega,$$

then we say $A_1, A_2, \ldots$ form a partition of Ω.

$$\left|A_1 \;\middle|\; A_2 \;\middle|\; A_3\right|_{\Omega}$$

Definition 6.1.6

$A, B,\ \mathcal{A} = \{A, B\}$. $\mathcal{A}$ — collection of subsets of Ω.

Example.

Suppose $\Omega = \{0, 1, 2\}$. It's clearly finite. What are *all* possible subsets of Ω?

$$\mathcal{P}(\Omega) = \{\varnothing, \{0\}, \{1\}, \{2\}, \{0, 1\}, \{0, 2\}, \{1, 2\}, \{0, 1, 2\}\}$$

where $\mathcal{P}$ is power set.

$$2^{|\Omega|} = (1+1)^3 = 2^3 = \binom{3}{0} + \binom{3}{1} + \binom{3}{2} + \binom{3}{3}.$$

6.2 The Probability Space

Probability Space:

$$(\underbrace{\Omega}_{\text{Abstract space}}, \underbrace{\mathcal{F}}_{\text{“suitable” collection of subsets of } \Omega}, \underbrace{\mathbb{P}}_{\text{probability measure}})$$

6.2.1 The σ-algebra

Definition 6.2.1

The set of all possible outcomes of an experiment is called the *sample space* and is denoted by Ω.

Definition 6.2.2

Let Ω be a set. A collection $\mathcal{F}$ of subsets of Ω is called an *algebra* if

1. $\Omega \in \mathcal{F}$.
2. $A \in \mathcal{F} \implies A^c \in \mathcal{F}$.
3. $A, B \in \mathcal{F} \implies A \cup B \in \mathcal{F}$.

Note that by property (2), property (1) is equivalent to $\varnothing \in \mathcal{F}$.

We see from the properties of an algebra that it is closed under finite unions (and hence finite intersections). However, it often occurs that the state space is infinite, in which closure under finite unions is insufficient and we require closure under infinite unions.

Definition 6.2.3

Let Ω be a set with associated σ-algebra $\mathcal{F}$. An *event* $A \in \mathcal{F}$ is a subset of Ω.

"Performing the experiment", we get an outcome $w \in \Omega$. If $w \in A$, we say A occurs; otherwise, A does not occur. If Ω is finite or countable infinite, we can take $\mathcal{F} = \mathcal{P}(\Omega)$ (powerset of Ω).

Definition 6.2.4

We call a non-empty collection of sets $\mathcal{F}$ a *monotone class* if

1. $A_1, A_2, \dots \in \mathcal{F}$ are such that $A_n \subseteq A_{n+1}$ for all $n \in \mathbb{N}_{>0}$, then $\displaystyle\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$.
2. $A_1, A_2, \dots \in \mathcal{F}$ are such that $A_n \supseteq A_{n+1}$ for all $n \in \mathbb{N}_{>0}$, then $\displaystyle\bigcap_{n=1}^{\infty} A_n \in \mathcal{F}$.

Definition 6.2.5

Let Ω be a set. A collection $\mathcal{F}$ of subsets of Ω is called a *σ-algebra* if

1. $\Omega \in \mathcal{F}$.
2. $A_1, A_2, \dots \in \mathcal{F} \implies \displaystyle\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$.
3. $A \in \mathcal{F} \implies A^c \in \mathcal{F}$.

To express the set Ω and its associated σ-algebra $\mathcal{F}$ succinctly, we use $(\Omega, \mathcal{F})$.

Note that $\mathcal{F} = \{\varnothing, \Omega\}$ is a σ-algebra; indeed, it is the smallest σ-algebra associated with Ω and is known as the trivial σ-algebra. *Similarly, the smallest σ-algebra of Ω to contain some $A \subseteq \Omega$ is $\mathcal{F} = \{\varnothing, A, A^c, \Omega\}$.*

σ-algebra $\mathcal{F}$ is an algebra.

Theorem 6.2.6

Let Ω be a set. Then $\mathcal{F}$ being both an algebra of subsets of Ω and a monotone class $\Leftrightarrow$ $\mathcal{F}$ being a σ-algebra of subsets of Ω.

Proof. ($\Leftarrow$) If $\mathcal{F}$ is a σ-algebra then clearly $\mathcal{F}$ is also an algebra. Furthermore, $\mathcal{F}$ satisfies property (1)

of a monotone class by definition. For property (2), observe that

$$\bigcap_{n=1}^{\infty} A_n = \left(\bigcup_{n=1}^{\infty} A_n^c \right)^c$$

by De Morgan's Laws, which are in $\mathcal{F}$ by properties (2) and (3) of a σ-algebra.

($\Rightarrow$) Properties (1) and (3) of a σ-algebra come directly from $\mathcal{F}$ being an algebra. For property (2), if $A_1, A_2 \in \mathcal{F}$ then $A_1 \cup A_2 \in \mathcal{F}$ since $\mathcal{F}$ is an algebra. Repeated application of this rule gets us $\bigcup_{i=1}^n A_i \in \mathcal{F}$ for $A_1, \ldots, A_n \in \mathcal{F}$. Define $B_n := \bigcup_{i=1}^n A_i$ and note that $B_1, B_2, \cdots \in \mathcal{F}$ with $B_1 \subseteq B_2 \subseteq \cdots$. Then by $\mathcal{F}$ being a monotone class we get

$$\bigcup_{n=1}^{\infty} B_n \in \mathcal{F} \qquad \text{where} \qquad \bigcup_{n=1}^{\infty} B_n = \bigcup_{n=1}^{\infty} \bigcup_{i=1}^{n} A_i = \bigcup_{n=1}^{\infty} A_n.$$

□

Definition 6.2.7

Let $\mathcal{A}$ be an arbitrary family of subsets of some set Ω. Then there exists a unique smallest σ-algebra which contains every set in $\mathcal{A}$. It is, in fact, the intersection of all σ-algebras (on Ω) containing $\mathcal{A}$. This σ-algebra is denoted $\sigma(\mathcal{A})$ and is called the *σ-algebra generated by $\mathcal{A}$* (in this course, we call it the *minimal* σ-algebra generated by $\mathcal{A}$).

Example.

> Let Ω be our state space, $A, B \subset \Omega$ be not disjoint and $\mathcal{A} = \{A, B\}$.

Then

$$\sigma(\mathcal{A}) = \big\{\varnothing, A, B, A \setminus B, A \cap B, B \setminus A, A \cup B, (A \cup B)^c, (A \cap B)^c, (A \setminus B)^c, (B \setminus A)^c,$$

$$(A \setminus B) \cup (B \setminus A), (A \setminus B) \cup (A \cup B)^c, (A \cap B) \cup (A \cup B)^c, (B \setminus A) \cup (A \cup B)^c\big\}.$$

If we had $\mathcal{A} = \{A, B, C\}$ with none of A, B and C disjoint, $|\sigma(\mathcal{A})| = 2^8 = 256$. On the other hand, if A, B and C were disjoint, $|\sigma(\mathcal{A})| = 2^4 = 16$.

Theorem 6.2.8

Let Ω be a set. If $\mathcal{F}$ and $\mathcal{G}$ are both σ-algebras of subsets of Ω, then $\mathcal{H} = \mathcal{F} \cap \mathcal{G}$ is also a σ-algebra of subsets of Ω. Moreover, if $\mathcal{T}$ is an index set and, $\forall\, t \in \mathcal{T}$, $\mathcal{F}_t$ is a σ-algebra of subsets of Ω, then $\bigcap_{t \in \mathcal{T}} \mathcal{F}_t$ is also a σ-algebra of subsets of Ω.

Example.

> *Borel σ-algebra*

Let $\Omega = \mathbb{R}$. The Borel σ-algebra on $\mathbb{R}$ is defined to be

$$\mathcal{B}(\mathbb{R}) := \sigma\big((-\infty, a) : a \in \mathbb{R}\big).$$

From this definition, one can work out that the following interval forms are all in $\mathcal{B}(\mathbb{R})$ (where $a, b \in \mathbb{R}$):

$$(-\infty, a), \quad (-\infty, a], \quad (a, \infty), \quad \{a\}, \quad (a, b), \quad (a, b], \quad [a, b), \quad [a, b].$$

One can similarly define the Borel σ-algebra over the extended real line $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, \infty\}$ to be $\overline{\mathcal{B}} = \mathcal{B}(\overline{\mathbb{R}})$.

Finally, one can define the Borel σ-algebra over a subset $A \subseteq \mathbb{R}$ via $\mathcal{B}(A) = \mathcal{B}(\mathbb{R}) \cap A$.

6.2.2 The Probability Measure

Definition 6.2.9: Kolmogorov Axioms

A *probability measure* $\mathbb{P}$ on $(\Omega, \mathcal{F})$ is a function $\mathbb{P} : \mathcal{F} \to [0, 1]$ satisfying

1. $\mathbb{P}(\varnothing) = 0$ and $\mathbb{P}(\Omega) = 1$.
2. If $A_1, A_2, \ldots$ are mutually disjoint members of $\mathcal{F}$, then

$$\mathbb{P}\Big(\bigcup_{i=1}^{\infty} A_i\Big) = \sum_{i=1}^{\infty} \mathbb{P}(A_i).$$

The triple $(\Omega, \mathcal{F}, \mathbb{P})$ is called a *probability space*.

Lemma 6.2.10

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space with $A, B \in \mathcal{F}$. Then

1. $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.
2. If $B \supseteq A$, then $\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A) \geq \mathbb{P}(A)$.
3. $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.
4. **Boole's Inequality.** For $A_1, A_2, \cdots \in \mathcal{F}$,

$$\mathbb{P}\Big(\bigcup_{n=1}^{\infty} A_n\Big) \leq \sum_{n=1}^{\infty} \mathbb{P}(A_n).$$

Example.

Coin Toss: Let $\Omega = \{H, T\}$ and $\mathcal{F} = \mathcal{P}(\Omega) = \{\varnothing, H, T, \Omega\}$.

Define $\mathbb{P}(\{H\}) := p$, which leads to $\mathbb{P}(\{T\}) = 1 - p$. For $A \in \mathcal{F}$ we have $\mathbb{P}(A) = \sum_{\omega \in A} \mathbb{P}(\{\omega\})$. For instance,

$$\mathbb{P}(\Omega) = \mathbb{P}(\{H\}) + \mathbb{P}(\{T\}) = p + 1 - p = 1.$$

Example.

Let $\Omega = \{a_1, \ldots, a_n\}$, $\mathcal{F} = \mathcal{P}(\Omega)$ and $\mathbb{P}(\{a_i\}) = p_i$ for all $i \in \{1, \ldots, n\}$.

Then for $A \in \mathcal{F}$,

$$\mathbb{P}(A) = \sum_{a_i \in A} \mathbb{P}(\{a_i\}).$$

Example.

Let $\Omega = [0, 2]$ and $\mathcal{F} = B([0, 2]) = \mathcal{B}(\mathbb{R}) \cap [0, 2]$.

Then for $A \in \mathcal{F}$,

$$\mathbb{P}(A) = \frac{\lambda(A)}{2},$$

where λ is the Lebesgue measure, which is our "usual notion of length".

Theorem 6.2.11: Continuity of probability measures

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $A_1, A_2, \dots$ be an increasing sequence of events, so that $A_1 \subseteq A_2 \subseteq A_3 \subseteq \cdots$. Then

$$\lim_{n\to\infty} \mathbb{P}(A_n) = \mathbb{P}\Big(\bigcup_{n=1}^{\infty} A_n\Big) = \mathbb{P}\Big(\lim_{n\to\infty} A_n\Big).$$

Similarly, if $B_1, B_2, \dots \in \mathcal{F}$ are such that $B_1 \supseteq B_2 \supseteq B_3 \supseteq \cdots$, then

$$\lim_{n\to\infty} \mathbb{P}(B_n) = \mathbb{P}\Big(\bigcap_{n=1}^{\infty} B_n\Big) = \mathbb{P}\Big(\lim_{n\to\infty} B_n\Big).$$

6.2.3 Borel-Cantelli Lemmas

Definition 6.2.12

Let $A_1, A_2, \dots$ be a countably infinite sequence of events. We shall use $\{A_n \text{ i.o.}\}$ to denote an event occurring for infinitely many indices n and $\{A_n \text{ a.b.f.m}\}$ to denote an event occurring for all but finitely many indices n. Then define

$$\limsup_n A_n := \bigcap_{m\geq 1} \bigcup_{n\geq m} A_n = \{A_n \text{ i.o.}\}, \qquad \liminf_n A_n := \bigcup_{m\geq 1} \bigcap_{n\geq m} A_n = \{A_n \text{ a.b.f.m}\}.$$

- $\liminf_n A_n = \Big(\limsup_n A_n^c\Big)^c$
- $\{A_n \ a.b.f.m\} \subseteq \{A_n \ i.o.\} \iff \liminf_n A_n \subseteq \limsup_n A_n.$

Example.

First think of a sequence of numbers, $a_1 = 2,\ a_2 = 1,\ a_3 = 2, \dots$

$$\limsup_n a_n = 2, \qquad \liminf_n a_n = 1.$$

Example.

$A_1 = [0,2],\ A_2 = [0,1],\ A_3 = [0,2], \dots$

$$\limsup_n A_n = [0,2], \qquad \liminf_n A_n = [0,1].$$

Definition 6.2.13

A family of events $\{A_i\}_{i\in I}$ for an indexing set I is called *independent* or *mutually independent* if

$$\mathbb{P}\Big(\bigcap_{i\in J} A_i\Big) = \prod_{i\in J} \mathbb{P}(A_i), \qquad \forall\, J \subseteq I,$$

where J is finite. If instead $\{A_i\}_{i\in I}$ is such that $\mathbb{P}(A_i \cap A_j) = \mathbb{P}(A_i)\mathbb{P}(A_j)$ for all $i \neq j$, it is called *pairwise independent*.

Lemma 6.2.14: Borel–Cantelli Lemmas

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space with $A_1, A_2, \dots \in \mathcal{F}$.

1. If $\sum_{n\geq 1} \mathbb{P}(A_n) < \infty$, then $\mathbb{P}\big(\{A_n \text{ i.o.}\}\big) = 0$.
2. If $\{A_n\}_{n=1}^{\infty}$ is mutually independent and $\sum_{n\geq 1} \mathbb{P}(A_n) = \infty$, then $\mathbb{P}\big(\{A_n \text{ i.o.}\}\big) = 1$.

Example.

Suppose $\mathbb{P}(A_n) = p_n$. If $p_n = \dfrac{1}{n^2}$, then

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} < \infty \implies \mathbb{P}(A_n \text{ i.o.}) = 0.$$

Example.

If instead $\mathbb{P}(A_n) = \dfrac{1}{n}$ and the $\{A_n\}$ are mutually independent, then

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \sum_{n=1}^{\infty} \frac{1}{n} = +\infty \implies \mathbb{P}(A_n \text{ i.o.}) = 1.$$

Proof. (1) Define

$$B_m = \bigcup_{n=m}^{\infty} A_n \qquad (m \geq 1),$$

so that $B_1 \supseteq B_2 \supseteq \cdots$. By Boole's Inequality,

$$\mathbb{P}(B_m) \leq \sum_{n=m}^{\infty} \mathbb{P}(A_n),$$

hence $\mathbb{P}(B_m) \to 0$ as $m \to \infty$. Moreover,

$$\mathbb{P}\big(\{A_n \text{ i.o.}\}\big) = \mathbb{P}\Big(\limsup_n A_n\Big) = \mathbb{P}\Big(\bigcap_{m=1}^{\infty} B_m\Big) = \lim_{N\to\infty} \mathbb{P}(B_N) = 0,$$

where the last equality follows from continuity from above.
(2) First observe that

$$\mathbb{P}\big(\{A_n^c \text{ a.b.f.m.}\}\big) = \mathbb{P}\Big(\liminf_n A_n^c\Big) = \mathbb{P}\Big(\bigcup_{m=1}^{\infty} \bigcap_{n=m}^{\infty} A_n^c\Big) = 1 - \mathbb{P}\Big(\limsup_n A_n\Big) = 1 - \mathbb{P}\big(\{A_n \text{ i.o.}\}\big).$$

For all $x \geq 0$, $1 - x \leq e^{-x}$. Since $\{A_n\}$ is mutually independent, Lemma 6.3.2 implies that $\{A_n^c\}$ is as well; for $M > m$,

$$\mathbb{P}\Big(\bigcap_{n=m}^{M} A_n^c\Big) = \prod_{n=m}^{M} \mathbb{P}(A_n^c) \leq \prod_{n=m}^{M} e^{-\mathbb{P}(A_n)} = \exp\Big(-\sum_{n=m}^{M} \mathbb{P}(A_n)\Big).$$

Therefore

$$0 \leq \lim_{M\to\infty} \mathbb{P}\Big(\bigcap_{n=m}^{M} A_n^c\Big) \leq \lim_{M\to\infty} \exp\Big(-\sum_{n=m}^{M} \mathbb{P}(A_n)\Big) = 0,$$

so

$$\mathbb{P}\Big(\bigcap_{n=m}^{\infty} A_n^c\Big) = 0 \quad (m \geq 1).$$

By Boole's Inequality,

$$0 \leq \mathbb{P}\big(\{A_n^c \text{ a.b.f.m.}\}\big) = \mathbb{P}\Big(\bigcup_{m=1}^{\infty}\bigcap_{n=m}^{\infty} A_n^c\Big) \leq \sum_{m=1}^{\infty}\mathbb{P}\Big(\bigcap_{n=m}^{\infty} A_n^c\Big) = 0.$$

Consequently,

$$0 = \mathbb{P}\big(\{A_n^c \text{ a.b.f.m.}\}\big) = 1 - \mathbb{P}\big(\{A_n \text{ i.o.}\}\big) \implies \mathbb{P}\big(\{A_n \text{ i.o.}\}\big) = 1.$$

□

6.3 Random Variables

6.3.1 The Random Variable and its Distribution

Definition 6.3.1

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A *random variable* is a function $X : \Omega \to \mathbb{R}$ with the property (in our context) that $\forall B \in \mathcal{B}(\mathbb{R}), \quad X^{-1}(B) := \{\, w \in \Omega : X(w) \in B \,\} \in \mathcal{F}$. Such a function is said to be $\mathcal{F}$*-measurable*.

Similarly, an "extended" random variable is a function $X : \Omega \to \overline{\mathbb{R}}$; a random variable is called improper *if it has a positive probability of taking on the values $\pm\infty \in \overline{\mathbb{R}}$. Otherwise, it is* proper.

Definition 6.3.2

A collection of random variables $X := (X_t)_{t\in T}$, where T is any index set, is called a *stochastic process*. When T is finite, X is called a *random vector*.

Hence, a random vector is a function $X : \Omega \to \mathbb{R}^d$, where $d = |T|$.

Definition 6.3.3

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and X a random variable within this space. The function

$$\mu_X : \mathcal{B}(\mathbb{R}) \to [0,1], \qquad B \mapsto \mathbb{P}\big(X^{-1}(B)\big),$$

is called the *distribution* of X.

By abuse of notation, we more commonly denote this as

$$\mu_X(B) = \mathbb{P}\big(X^{-1}(B)\big) = \mathbb{P}\big(\{\, w \in \Omega : X(w) \in B \,\}\big) = \mathbb{P}(X \in B).$$

Recall that previously, we introduced the cdf $F_X : \mathbb{R} \to [0,1]$ as $F_X(x) = \mathbb{P}(X \leq x)$, $x \in \mathbb{R}$. What this really means, using the above notation, is $\mathbb{P}(X \in (-\infty, x])$, i.e. $F_X(x) = \mu_X((-\infty, x])$, $x \in \mathbb{R}$.

Example.

> Consider the probability space $\big([0,1], \mathcal{B}([0,1]), \lambda\big)$, where λ is the Lebesgue measure. Then
>
> $$X(w) = \mathbf{1}_{[0,p]}(w) = \begin{cases} 1, & w \in [0,p], \\ 0, & \text{otherwise.} \end{cases}$$

To find the distribution of X, there are three cases to consider:

1. $\forall x < 0,\ X^{-1}\big((-\infty, x]\big) = \varnothing$, so $F_X(x) = \mathbb{P}(\varnothing) = 0$.

2. $\forall 0 \le x < 1,\ X^{-1}\big((-\infty, x]\big) = [p, 1]$, so $F_X(x) = \mathbb{P}([p, 1]) = 1 - p$.
3. $\forall x \ge 1,\ X^{-1}\big((-\infty, x]\big) = [0, 1]$, so $F_X(x) = \mathbb{P}([0, 1]) = 1$.

In other words, we have shown that $X \sim \mathrm{Ber}(p)$.

Example.

> Consider the probability space $\big([0,1]^2, \mathcal{B}([0,1]^2), \lambda\big)$, where λ is the Lebesgue measure (for this example, it just means "area").

Logic: $X \to X^{-1} \in \mathcal{F} \to \mathbb{P}$

For all $w \in \Omega$, write $w := (w_1, w_2)$ and for the random vector define

$$\mathbf{X}(w) := \begin{pmatrix} X_1(w) \\ X_2(w) \end{pmatrix} = \begin{pmatrix} \mathbf{1}_{[0,p]}(w_1) \\ \mathbf{1}_{[0,p]}(w_2) \end{pmatrix}.$$

In other words, $\mathbf{X}$ maps

$$w \in [0,p]^2 \to (1,1), \quad w \in [0,p] \times (p,1] \to (1,0), \quad w \in (p,1] \times [0,p] \to (0,1), \quad w \in (p,1]^2 \to (0,0).$$

Recall that

$$F_{\mathbf{X}}(x_1, x_2) = \mathbb{P}(X_1 \le x_1,\ X_2 \le x_2) = \mathbb{P}\Big(\begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \in (-\infty, x_1] \times (-\infty, x_2]\Big) = \mathbb{P}\big(\{\, w \in \Omega : \mathbf{X}(w) \in (-\infty, x_1] \times (-\infty, x_2] \,\}\big).$$

There are five cases to consider for $\Gamma := (-\infty, x_1] \times (-\infty, x_2]$.

1. $(x_1 < 0) \vee (x_2 < 0)$. $X^{-1}(\Gamma) = \varnothing$, so $\mathbb{P}(X^{-1}(\Gamma)) = 0$.
2. $0 \le x_1, x_2 < 1$. $X^{-1}(\Gamma) = (p,1]^2$, so $\mathbb{P}(X^{-1}(\Gamma)) = (1-p)^2$.
3. $(0 \le x_1 < 1) \wedge (x_2 \ge 1)$. $X^{-1}(\Gamma) = (p,1]^2 \cup [0,p] \times [0,p]$, so
$$\mathbb{P}(X^{-1}(\Gamma)) = \lambda((p,1]^2) + \lambda([0,p] \times [0,p]) = (1-p)^2 + p(1-p) = 1 - p.$$
4. $(x_1 \ge 1) \wedge (0 \le x_2 < 1)$. $X^{-1}(\Gamma) = (p,1]^2 \cup [0,p] \times (p,1]$, so
$$\mathbb{P}(X^{-1}(\Gamma)) = \lambda((p,1]^2) + \lambda([0,p] \times (p,1]) = (1-p)^2 + p(1-p) = 1 - p.$$
5. $(x_1 \ge 1) \wedge (x_2 \ge 1)$. $X^{-1}(\Gamma) = [0,1]^2$, so $\mathbb{P}(X^{-1}(\Gamma)) = 1$.

In summary,

$$F_{\mathbf{X}}(x_1, x_2) = \mathbb{P}(X_1 \le x_1,\ X_2 \le x_2) = \begin{cases} 0, & (x_1 < 0) \vee (x_2 < 0), \\ (1-p)^2, & 0 \le x_1, x_2 < 1, \\ 1 - p, & (0 \le x_1 < 1) \wedge (x_2 \ge 1), \\ 1 - p, & (x_1 \ge 1) \wedge (0 \le x_2 < 1), \\ 1, & (x_1 \ge 1) \wedge (x_2 \ge 1). \end{cases}$$

We see clearly from the previous example that $X_1, X_2 \sim \mathrm{Ber}(p)$, so

$$F_{X_1}(x) = F_{X_2}(x) = \begin{cases} 0, & x < 0, \\ 1 - p, & 0 \le x < 1, \\ 1, & x \ge 1. \end{cases}$$

One can now verify by direct computation that $F_{\mathbf{X}}(x_1, x_2) = F_{X_1}(x_1) F_{X_2}(x_2)$, i.e. that X_1 and X_2 are independent.

Theorem 6.3.4: Carathéodory's Extension Theorem

Let X be a random variable in a probability space and suppose $\mu_X \to F_X$ (i.e. the measure μ_X corresponds to the cdf F_X). Given F_X, $\exists(\Omega, \mathcal{F}, \mathbb{P})$ a probability space and a random variable $\tilde{X}$ such that

$$\mu_{\tilde{X}}\big((-\infty, x]\big) = F_X(x), \qquad \forall x \in \mathbb{R}.$$

every cdf has a corresponding random variable.

Theorem 6.3.5: Kolmogorov's Extension Theorem

Given mutually consistent but otherwise arbitrary joint cdfs, $\exists(\Omega, \mathcal{F}, \mathbb{P})$ a probability space and a countable sequence of random variables $\{X_n\}$ with said joint cdfs.

every joint cdf has a corresponding a countable sequence of random variables.

Given a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and a random vector $\mathbf{X} : \Omega \to \mathbb{R}^d$, we say $\mathbf{X}$ induces a distribution $\mu_{\mathbf{X}} : \mathcal{B}(\mathbb{R}^d) \to [0, 1]$ via

$$\mu_{\mathbf{X}}(B) = \mathbb{P}(\mathbf{X} \in B), \qquad \forall B \in \mathcal{B}(\mathbb{R}^d).$$

Theorem 6.3.6

Given a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and a random variable $X : \Omega \to \mathbb{R}$, the cumulative distribution function (cdf) of X, F, defined via $F(x) := \mathbb{P}(X \leq x)$ for all $x \in \mathbb{R}$, satisfies the following properties:

1. $\forall x \in \mathbb{R},\ F(x) \in [0, 1]$.
2. $x \leq y \implies F(x) \leq F(y)$.
3. If $x_n \to x^-$ (i.e. x_n approaches x from below), then $F(x_n) \to F(x^-) = \mathbb{P}(X < x)$.
4. If $x_n \to x^+$ (i.e. x_n approaches x from above), then $F(x_n) \to F(x^+) = \mathbb{P}(X \leq x)$.
5. F is right-continuous, i.e. $F(x + h) \to F(x)$ as $h \to 0^+$.

Theorem 6.3.7: Lebesgue Decomposition Theorem

Every cdf $F(x)$ can be decomposed as

$$F(x) = \alpha F_d(x) + \beta F_{\text{a.c.}}(x) + (1 - \alpha - \beta) F_{\text{s.c.}}(x), \qquad (\alpha, \beta \geq 0) \wedge (\alpha + \beta \leq 1),$$

where

- F_d is the cdf of a discrete random variable;
- $F_{\text{a.c.}}$ is the cdf of an absolutely continuous random variable, i.e. one for which

$$f(x) = \frac{\mathrm{d}}{\mathrm{d}x} F_{\text{a.c.}}(x) \quad \text{such that} \quad \int_{-\infty}^{x} f(u)\,\mathrm{d}u = F_{\text{a.c.}}(x);$$

- $F_{\text{s.c.}}$ is the cdf of a singular continuous random variable, i.e. one for which the pdf is zero almost everywhere (meaning it is only non-zero on a set of points with measure zero).

6.3.2 Types of Random Variables

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. We present here some different types of random variables:

Definition 6.3.8: Indicator function.

Let $A \in \mathcal{F}$. Then we can define a random variable via

$$X(w) = \mathbf{1}_A(w) = \begin{cases} 1, & w \in A, \\ 0, & w \notin A. \end{cases}$$

Example.

$\big([0,1], \mathcal{B}([0,1]), \lambda\big)$ $X = \mathbf{1}_{\mathbb{Q}^c}$ or $\mathbf{1}_{[0,p]}$

Example.

$\big(\{1,\dots,6\}, \mathcal{P}(\Omega), \mathbb{P}\big) \quad X = \mathbf{1}_{\{\xi \text{ even}\}}$

Definition 6.3.9: Simple Random Variables.

For any finite $n \in \mathbb{N}_{\geq 1}$, let $A_1, \dots, A_n \in \mathcal{F}$ and $b_1, \dots, b_n \in \mathbb{R}$. Then a simple random variable is defined via

$$X_n(w) := \sum_{i=1}^{n} b_i \mathbf{1}_{A_i}(w).$$

Example.

$\big(\{1,\dots,6\}, \mathcal{P}(\Omega), \mathbb{P}\big)$, $\mathbb{P}(\{a\}) = \frac{1}{6}$, $a \in \Omega$.
$X = 1 \cdot \mathbf{1}_{\{\omega=1\}} + 2 \cdot \mathbf{1}_{\{\omega=2\}} + \cdots + 6 \cdot \mathbf{1}_{\{\omega=6\}}$ or $5 \cdot \mathbf{1}_{\{\xi \text{ even}\}} - 5 \cdot \mathbf{1}_{\{\xi \text{ odd}\}}$

Definition 6.3.10: Limits of simple random variables.

Let $A_1, A_2, \dots \in \mathcal{F}$ and $b_1, b_2, \dots \in \mathbb{R}$. Then we can define

$$X_\infty(w) = \sum_{i=1}^{\infty} b_i \mathbf{1}_{A_i}(w).$$

Example.

$$\big([0,1], \mathcal{B}([0,1]), \lambda\big), \qquad X(w) = \sum_{n=0}^{\infty} \frac{1}{2^n} \mathbf{1}_{[0,\, 2^{-n}]}(w)$$

Example.

$$\big(\mathbb{N}, \mathcal{P}(\mathbb{N}), \mathbb{P}\big), \qquad \mathbb{P}(\{\xi = a\}) = p^a(1-p), \quad a \in \mathbb{N}, \qquad X(w) = \sum_{i=0}^{\infty} \mathbf{1}_{[0,\, i]}(w)$$

Definition 6.3.11: Non-negative random variables.

A non-negative random variable is simply a function $X : \Omega \to \overline{\mathbb{R}}_{\geq 0}$ which satisfies the usual requirements for a random variable.

Example.

$$(0,1),\ \mathcal{B}((0,1),\lambda), \qquad X(w) = -\ln w \quad \text{or} \quad X(w) = 1 - \tfrac{1}{2}\log_2 w$$

Example.

$$([-1,1],\mathcal{B}([-1,1]),\lambda/2), \qquad X(w) = w^2.$$

Theorem 6.3.12

Any non-negative random variable X can be written as the pointwise (on the state space Ω) limit of a sequence of simple random variables $\{X_n\}$, i.e.

$$\lim_n X_n(w) = X(w), \qquad \forall\, w \in \Omega.$$

Example.

The pointwise limit of

$$X_n(w) := \begin{cases} \dfrac{k}{2^n}, & \dfrac{k}{2^n} \le X(w) \le \dfrac{k+1}{2^n}, \quad 0 \le k \le 2^n - 1, \\ n, & X(w) \ge n, \end{cases}$$

in the state space Ω is a non-negative random variable.

Definition 6.3.13: Arbitrary random variables.

A random variable $X : \Omega \to \overline{\mathbb{R}}$ such that

$$X^+(w) = \max\{X(w), 0\} = \begin{cases} X(w), & X(w) \ge 0, \\ 0, & \text{otherwise}, \end{cases}$$

$$X^-(w) = \max\{-X(w), 0\} = \begin{cases} -X(w), & X(w) \le 0, \\ 0, & \text{otherwise}. \end{cases}$$

We set $X(w) = X^+(w) - X^-(w)$ for all $w \in \Omega$ except if both $X^+(w) = X^-(w) = \infty$, in which case the random variable is undefined for these values of w.

Example.

$\big([0,1],\mathcal{B}([0,1]),\lambda\big)$ the random variable $X(w) = \Phi^{-1}(w)$ for all $w \in \Omega$, where Φ is the cdf of $\mathcal{N}(0,1)$

Example.

$\big((0,1)\times(0,1),\mathcal{B}((0,1)\times(0,1)),\lambda\big)$ the random variable

$$X(w) = \big(2\mathbf{1}_{\{w_1<1/2\}} - 1\big)\log(w_2), \qquad \forall (w_1,w_2) \in \Omega.$$

6.3.3 Expectation

Let $(\Omega,\mathcal{F},\mathbb{P})$ be a probability space and let X be a random variable. We introduce the expectation of a random variable in four steps:

1. If X is an indicator, i.e. $X = \mathbf{1}_A$ for some $A \in \mathcal{F}$, then we define

$$\mathbb{E}X = \mathbb{E}\mathbf{1}_A := \mathbb{P}(A).$$

2. If X is simple, i.e. for some finite $n \in \mathbb{N}_{\geq 1}$, $A_1, \ldots, A_n \in \mathcal{F}$ and $b_1, \ldots, b_n \in \mathbb{R}$, and $X = \sum_{i=1}^n b_i \mathbf{1}_{A_i}$, then we define
$$\mathbb{E}X = \mathbb{E}\Big(\sum_{i=1}^n b_i \mathbf{1}_{A_i}\Big) := \sum_{i=1}^n b_i \mathbb{E}\mathbf{1}_{A_i}.$$

3. If $X \geq 0$ then recall we may write it as a pointwise (on Ω) limit of a sequence $X_1, X_2, \ldots$ of simple random variables. That is, $X(w) = \lim_{n\to\infty} X_n(w)$ for all $w \in \Omega$, where $X_n = \sum_{i=1}^n b_i^{(n)} \mathbf{1}_{A_i^{(n)}}$. Hence, we define
$$\mathbb{E}X = \mathbb{E}\Big(\lim_{n\to\infty} X_n\Big) := \lim_{n\to\infty} \mathbb{E}X_n.$$

4. If X is arbitrary, then decompose $X = X^+ - X^-$ with X^+ and X^- non-negative. Then if $\mathbb{E}X^+ = \mathbb{E}X^- = \infty$, we say $\mathbb{E}X$ is undefined. Otherwise, we define
$$\mathbb{E}X = \mathbb{E}(X^+ - X^-) := \mathbb{E}X^+ - \mathbb{E}X^-.$$

Definition 6.3.14

Expectation defined in this way is written symbolically as
$$\mathbb{E}X = \int_\Omega X \,\mathrm{d}\mathbb{P}.$$

Theorem 6.3.15

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and X an arbitrary (proper) random variable. Furthermore, let $g : \mathbb{R} \to \mathbb{R}$ be a measurable function. Then
$$\mathbb{E}g(X) = \int_\Omega g(X)\,\mathrm{d}\mathbb{P} = \int_\mathbb{R} g(x)\,\mu_X(\mathrm{d}x) = \int_\mathbb{R} g(x)\,\mathrm{d}F_X(x),$$
where F_X is the cdf induced by $(\Omega, \mathcal{F}, \mathbb{P})$ and $X : \Omega \to \mathbb{R}$.

The last integral is called a **Lebesgue–Stieltjes integral**. Moreover, it reduces to simpler expressions in special cases:

1. If $F_X \equiv F_d$ (i.e. F_X is *purely discrete*) and $\mathbb{P}(X \in S)$ for a countable set S, then
$$\int_\mathbb{R} g(x)\,\mathrm{d}F_X(x) \;=\; \sum_{x\in S} g(x)\,\mathbb{P}\big(X = x\big).$$

2. If $F_X \equiv F_{\text{a.c.}}$ (i.e. F_X is *purely absolutely continuous*), then
$$\int_\mathbb{R} g(x)\,\mathrm{d}F_X(x) \;=\; \int_\mathbb{R} g(x)\,f_X(x)\,\mathrm{d}x,$$
where
$$f_X(x) \;=\; \frac{\mathrm{d}}{\mathrm{d}x} F_{\text{a.c.}}(x), \qquad \forall\, x \in \mathbb{R}.$$

Theorem 6.3.16

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, $a, b \in \mathbb{R}$, and X and Y random variables defined on this space. Then $\mathbb{E}$ has the following properties.

(1) *(Linearity).* $\mathbb{E}(aX + bY) = a\,\mathbb{E}X + b\,\mathbb{E}Y$.

(2) *(Monotonicity).* If $Y \geq X$ on $\mathbb{R}$, then $\mathbb{E}Y \geq \mathbb{E}X$.

(3) *(Absolute Integrability).* $\mathbb{E}X < \infty \iff \mathbb{E}|X| < \infty$.

(4) *(Jensen's Inequality).* For a convex function $\varphi : \mathbb{R} \to \mathbb{R}$, $\varphi\big(\mathbb{E}X\big) \leq \mathbb{E}\varphi(X)$.

6.3.4 Convergence

In step 3 of defining $\mathbb{E}$ above, we had

$$\mathbb{E}X = \mathbb{E}\lim_{n\to\infty} X_n := \lim_{n\to\infty} \mathbb{E}X_n$$

for a sequence of simple random variables $\{X_n\}$ converging pointwise (on Ω) to X. However, this limit equality does not hold for arbitrary sequences of random variables converging pointwise to X.

Example.

As a counter-example, consider $\big([0,1], \mathcal{B}([0,1]), \lambda\big)$ and define $X_n : \Omega \to \mathbb{R}$ via

$$X_n(w) = n\,e^{-nw}, \qquad w \in [0,1],\ n \in \mathbb{N}_{\geq 1}.$$

Proof. Now

$$X(w) = \lim_{n\to\infty} X_n(w) = \begin{cases} 0, & w \in (0,1], \\ \infty, & w = 0. \end{cases}$$

Hence

$$\mathbb{E}X = \mathbb{E}\lim_{n\to\infty} X_n = \infty \cdot \mathbb{P}(X = 0) + 0 \cdot \mathbb{P}\big(X \in (0,1]\big) = 0,$$

while

$$\mathbb{E}X_n = \int_\Omega X_n \,\mathrm{d}\mathbb{P} = \int_0^1 n\,e^{-nw}\,\mathrm{d}w = 1 - e^{-n} \implies \lim_{n\to\infty} \mathbb{E}X_n = 1 \neq 0.$$

□

Theorem 6.3.17: Monotone Convergence Theorem

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $X_1, X_2, \ldots$ be random variables on this space such that $0 \leq X_1 \leq X_2 \leq \cdots$ for almost all $w \in \Omega$ (i.e. for all but a subset of points with measure zero). Then $X = \lim_{n\to\infty} X_n$ exists and

$$\lim_{n\to\infty} \mathbb{E}X_n = \mathbb{E}\lim_{n\to\infty} X_n = \mathbb{E}X.$$

Example.

Suppose $Y_1, Y_2, \ldots$ is a sequence of non-negative random variables. Then, defining $X_n = \sum_{i=1}^n Y_i$, it holds via monotone convergence that

(1) $\displaystyle\sum_{i=1}^{\infty} Y_i$ exists (possibly ∞);

(2) $\mathbb{E}\sum_{i=1}^{\infty} Y_i = \sum_{i=1}^{\infty} \mathbb{E}Y_i$.

Lemma 6.3.18: Fatou Lemma

Let $X_1, X_2, \ldots$ be a sequence of non-negative random variables and define $X(w) = \liminf_{n\to\infty} X_n(w)$. Then

$$\mathbb{E}X = \mathbb{E}\liminf_{n\to\infty} X_n \leq \liminf_{n\to\infty} \mathbb{E}X_n.$$

Lemma 6.3.19: Reverse Fatou Lemma

Let $X_1, X_2, \ldots$ be an arbitrary sequence of random variables and Y be a non-negative random variable with $\mathbb{E}Y < \infty$ such that $X_n \leq Y$ for all $n \in \mathbb{N}_{\geq 1}$ for almost all $w \in \Omega$. Defining $X(w) = \limsup_{n\to\infty} X_n(w)$, it holds that

$$\limsup_{n\to\infty} \mathbb{E}X_n \leq \mathbb{E}\big(\limsup_{n\to\infty} X_n\big) = \mathbb{E}X.$$

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $X_1, X_2, \ldots$ be an arbitrary sequence of random variables on this space, along with the random variable X. Then recall that pointwise convergence means

$$\lim_{n\to\infty} X_n(w) = X(w), \quad \forall w \in \Omega.$$

Definition 6.3.20: Almost sure convergence

Almost sure convergence (i.e. $X_n \to X$ almost surely) means $\mathbb{P}\left(\lim_{n\to\infty} X_n = X\right) = 1$.

Theorem 6.3.21: Dominated Convergence Theorem

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $X_1, X_2, \ldots$ be an arbitrary sequence of random variables on this space. Moreover, let $|X_n| \leq Y$ almost surely for some random variable Y with $\mathbb{E}Y < \infty$ and assume $X = \lim_{n\to\infty} X_n$ almost surely. Then

$$\lim_{n\to\infty} \mathbb{E}X_n = \mathbb{E}\lim_{n\to\infty} X_n = \mathbb{E}X.$$

Example.

Let $X_n \sim \mathcal{U}[0, 1/n]$ for $n \in \mathbb{N}_{\geq 1}$ and $Y = 1$. Then $\mathbb{E}Y = 1 < \infty$, while $\mathbb{E}X_n = 1/(2n)$. Hence,

$$\mathbb{E}\lim_{n\to\infty} X_n = \lim_{n\to\infty} \mathbb{E}X_n = \lim_{n\to\infty} \frac{1}{2n} = 0.$$

Definition 6.3.22: Convergence in Probability

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $X_1, X_2, \ldots$ be an arbitrary sequence of random variables on this space, along with the random variable X. Then we say $X_n \to X$ ***in probability*** *if $\forall \varepsilon > 0$,*

$$\lim_{n\to\infty} \mathbb{P}(|X_n - X| \geq \varepsilon) = 0.$$

Example.

> We demonstrate in this example that the above mentioned forms of convergence are not equivalent. Let $X_n \sim \mathrm{Ber}(1/n)$ for $n \in \mathbb{N}_{\geq 1}$ be an independent sequence of Bernoulli trials and let $X = 0$.

For any $\varepsilon > 0$,

$$\mathbb{P}(|X_n - X| \geq \varepsilon) = \mathbb{P}(|X_n| \geq \varepsilon) = \mathbb{P}(X_n = 1) = \frac{1}{n} \to 0 \quad \text{as } n \to \infty.$$

Hence, $X_n \to X$ in probability. However, $\{X_n = 1\}$ are a sequence of independent events with $p_n := \mathbb{P}(X_n = 1) = 1/n$ and

$$\sum_{n=1}^{\infty} p_n = \sum_{n=1}^{\infty} \frac{1}{n} = \infty.$$

Therefore, by Borel-Cantelli,

$$\mathbb{P}\left(\limsup_{n\to\infty}\{X_n = 1\}\right) = 1 \implies \mathbb{P}\left(\lim_{n\to\infty} X_n = 0\right) = 0,$$

meaning X_n does not converge almost surely to $X = 0$.

Definition 6.3.23: Convergence in Distribution

Let $X_1, X_2, \ldots$ be a sequence of random variables with cdfs $F_1, F_2, \ldots$ and let X be a random variable with cdf F. If $\lim_{n\to\infty} F_n(x) = F(x)$ for all $x \in \mathbb{R}$ such that F is continuous, we say that X_n ***converges in distribution*** *to X.*

Convergence in distribution is also commonly called weak convergence, or "convergence in law".

Example.

> Convergence in distribution does not imply convergence in probability in general. Indeed, let $X \sim \mathcal{N}(0,1)$ and $X_n = -X$ for all $n \in \mathbb{N}_{\geq 1}$. Clearly, $X_n \sim \mathcal{N}(0,1)$ and $X_n \to X$ in distribution trivially. However, for any $\varepsilon > 0$,
>
> $$\mathbb{P}(|X_n - X| \geq \varepsilon) = \mathbb{P}(|-2X| \geq \varepsilon) = \mathbb{P}(X \geq \varepsilon/2) > 0,$$
>
> so X_n does not converge to X in probability.

Definition 6.3.24: Convergence in L_p-norm

Let $p \geq 1$ and $X_1, X_2, \ldots$ be a sequence of random variables such that $\mathbb{E}(|X_n|^p) < \infty$. Furthermore, let X be a random variable such that $\mathbb{E}(|X|^p) < \infty$. Then if

$$\lim_{n\to\infty} \mathbb{E}(|X_n - X|^p) = 0,$$

we say that X_n ***converges in L_p-norm*** *to X.*

Example.

> Convergence in probability does not necessarily imply convergence in L_p-norm.
> Consider $X_n = \sqrt{n}\,\mathbb{1}_{\{0 \leq U \leq 1/n\}}$, with $U \sim \mathcal{U}[0,1]$, and $X = 0$.

Then $\forall \varepsilon > 0$, we have

$$\mathbb{P}(|X_n - X| \geq \varepsilon) = \mathbb{P}\left(\sqrt{n}\,\mathbb{1}_{\{0 \leq U \leq 1/n\}} \geq \varepsilon\right) = \frac{1}{n}\,\mathbb{1}_{\{\sqrt{n} \geq \varepsilon\}} \to 0 \quad \text{as } n \to \infty.$$

Hence, $X_n \to X$ in probability. However,

$$\mathbb{E}(|X_n - X|^2) = \mathbb{E}\left(n\left(\mathbb{1}_{\{0\le U\le 1/n\}}\right)^2\right) = n\mathbb{P}(0 \le U \le 1/n) = n \cdot \frac{1}{n} = 1, \quad n \in \mathbb{N}_{\ge 1}.$$

Thus, X_n does not converge to X in L_2-norm.

In general, we have the following implications between the different modes of convergence:

$$\begin{tikzcd}
\text{pointwise} \arrow[r] & \text{almost surely} \arrow[d] & \\
& \text{in probability} \arrow[r] & \text{in distribution} \\
\text{in } L_q\text{-norm} \arrow[r, "q \ge p \ge 1"] & \text{in } L_p\text{-norm} \arrow[u] &
\end{tikzcd}$$

Additional restrictions/conditions are needed in general to have the reverse implications hold.

Theorem 6.3.25: Lévy's Continuity Theorem

Let $F_1, F_2, \ldots$ be a sequence of cdfs with corresponding characteristic functions $\phi_1, \phi_2, \ldots$.

1. *If $F_n \to F$ in distribution for some cdf F with characteristic function ϕ, then $\forall t \in \mathbb{R}$, $\phi_n(t) \to \phi(t)$ as $n \to \infty$.*
2. *If $\phi(t) = \lim_{n\to\infty} \phi_n(t)$ for all $t \in \mathbb{R}$ and $\phi(t)$ is continuous at $t = 0$, then ϕ is a characteristic function and, if F is its corresponding cdf, $F_n \to F$ in distribution.*

The characteristic function ϕ of a random variable X is the map $t \mapsto \mathbb{E}e^{itX}$, $t \in \mathbb{R}$.

In the above theorem, we wrote $F_n \to F$ in distribution, which is the weak convergence of functions, that is, pointwise convergence of a sequence of functions to another function at all continuity points. This is essentially the same as convergence in distribution of a sequence of random variables $X_1, X_2, \ldots$ to X with corresponding cdfs $F_1, F_2, \ldots$ and F. Recall that the extension theorems guaranteed that there is some underlying probability space $(\Omega, \mathcal{F}, \mathbb{P})$ on which $X_1, X_2, \ldots$ and X can be defined such that they have cdfs $F_1, F_2, \ldots$ and F, respectively.

6.3.5 Law of Large Numbers

Recall.

1. (Markov's Inequality). For any non-negative random variable $X \ge 0$ and any constant $a > 0$,
$$\mathbb{P}(X \ge a) \le \frac{\mathbb{E}X}{a}.$$
2. (Chebyshev Inequality). For any random variable X with $\sigma = \sqrt{\mathrm{Var}(X)} < \infty$, writing $\mu = \mathbb{E}X$, for any $a > 0$, we have
$$\mathbb{P}(|X - \mu| \ge a\sigma) \le \frac{1}{a^2}.$$

Theorem 6.3.26: Weak Law of Large Numbers

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X_1, X_2, \ldots$ be an iid sequence of random variables in this space with common mean $\mu = \mathbb{E}X_i < \infty$. Defining $S_n := \sum_{i=1}^n X_i$ for $n \in \mathbb{N}_{\ge 1}$, it holds that $S_n/n \to \mu$ in probability.

Proof. For this proof, we assume additionally that $\sigma^2 = \mathrm{Var}(X_i) < \infty$. Let $T_n = S_n/n - \mu$ for $n \in \mathbb{N}_{\ge 1}$. Then

$$\mathbb{E}T_n = \frac{1}{n} \cdot n\mu - \mu = 0 \quad \Longrightarrow \quad \mathbb{E}\frac{S_n}{n} = \mu, \quad n \in \mathbb{N}_{\ge 1}.$$

Moreover,

$$\mathrm{Var}(T_n) = \mathrm{Var}\left(\frac{S_n}{n} - \mu\right) = \frac{1}{n^2} \cdot n\sigma^2 = \frac{\sigma^2}{n}.$$

By Chebyshev,

$$\mathbb{P}(|T_n| \geq \varepsilon) = \mathbb{P}\left(\left|\frac{S_n}{n} - \mu\right| \geq \varepsilon\right) \leq \frac{\sigma^2}{n\varepsilon^2},$$

which goes to 0 as $n \to \infty$ for any $\varepsilon > 0$. Therefore,

$$\lim_{n\to\infty} \mathbb{P}\left(\left|\frac{S_n}{n} - \mu\right| \geq \varepsilon\right) = 0, \quad \forall \varepsilon > 0,$$

meaning $S_n/n \to \mu$ in probability. □

Theorem 6.3.27: Strong Law of Large Numbers

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X_1, X_2, \ldots$ be an iid sequence of random variables in this space with common mean $\mu = \mathbb{E}X_i < \infty$. Defining $S_n := \sum_{i=1}^n X_i$ for $n \in \mathbb{N}_{\geq 1}$, it holds that $S_n/n \to \mu$ almost surely. □

Proof. We again assume additionally that $\sigma^2 = \mathrm{Var}(X_i) < \infty$. The proof proceeds in two stages: (1) Assume $\mu_4 = \mathbb{E}(X_i^4) < \infty$, and (2) Relax to $\sigma^2 < \infty$, and without loss of generality, $\mu = 0$.

(1) By Markov's Inequality, we have

$$\mathbb{P}\left(\left|\frac{S_n}{n} - 0\right| \geq \varepsilon\right) \leq \frac{\mathbb{E}\left(\left(\frac{S_n}{n}\right)^4\right)}{\varepsilon^4} = \frac{\mathbb{E}(S_n^4)}{n^4\varepsilon^4}.$$

Now,

$$\mathbb{E}(S_n^4) = \cdots = n\mu_4 + 3n(n-1)\sigma^4 \leq n\mu_4 + 3n^2\sigma^4,$$

meaning

$$\mathbb{P}\left(\left|\frac{S_n}{n} - 0\right| \geq \varepsilon\right) \leq \frac{n\mu_4 + 3n^2\sigma^4}{n^4\varepsilon^4} = \frac{\mu_4}{n^3\varepsilon^4} + \frac{3\sigma^4}{n^2\varepsilon^4}.$$

Define events $A_n(\varepsilon) := \{|S_n/n - 0| \geq \varepsilon\}$ and observe that $\forall \varepsilon > 0$,

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n(\varepsilon)) < \infty,$$

which implies, by Borel-Cantelli, that

$$\mathbb{P}\left(\{A_n(\varepsilon) \text{ i.o.}\}\right) = 0.$$

Therefore, $S_n/n \to 0$ almost surely.

(2). Define $B_k(\varepsilon) := \{|S_k/k - 0| \geq \varepsilon\}$. Now by Chebyshev,

$$\mathbb{P}(B_{n^2}(\varepsilon)) = \mathbb{P}\left(\left|\frac{S_{n^2}}{n^2} - 0\right| \geq \varepsilon\right) \leq \frac{1}{\varepsilon^2}\mathrm{Var}\left(\frac{S_{n^2}}{n^2}\right) = \frac{n^2\sigma^2}{n^4\varepsilon^2} = \frac{\sigma^2}{n^2\varepsilon^2}.$$

By Borel-Cantelli,

$$\sum_{n=1}^{\infty} \mathbb{P}(B_{n^2}(\varepsilon)) < \infty \quad \Longrightarrow \quad \mathbb{P}\left(\{B_{n^2}(\varepsilon) \text{ i.o.}\}\right) = 0.$$

Thus, $S_{n^2}/n^2 \to 0$ almost surely. Define

$$D_n := \max_{n^2 \leq k < (n+1)^2} |S_k - S_{n^2}|.$$

Then

$$D_n^2 = \max_{n^2 \le k < (n+1)^2} (S_k - S_{n^2})^2 \le \sum_{k=n^2}^{(n+1)^2-1} (S_k - S_{n^2})^2,$$

and so

$$\mathbb{E}(D_n^2) \le \sum_{k=n^2}^{(n+1)^2-1} \mathbb{E}\left((S_k - S_{n^2})^2\right) = \sum_{k=n^2}^{(n+1)^2-1} (k - n^2)\sigma^2 \le \sum_{k=n^2}^{(n+1)^2-1} 2n\sigma^2 = 4n^2\sigma^2.$$

By Markov's Inequality, the above implies that

$$\mathbb{P}\left(\frac{D_n}{n^2} \ge \varepsilon\right) \le \frac{4n^2\sigma^2}{(n^2\varepsilon)^2} = \frac{4\sigma^2}{n^2\varepsilon^2}, \quad \forall \varepsilon > 0,$$

meaning $D_n/n^2 \to 0$ almost surely. Hence, for $n^2 \le k < (n+1)^2$, we have

$$\left|\frac{S_k}{k} - 0\right| = \left|\frac{S_{n^2} + (S_k - S_{n^2})}{k}\right| \le \frac{|S_{n^2}|}{k} + \frac{|S_k - S_{n^2}|}{k} \le \frac{|S_{n^2}|}{n^2} + \frac{D_n}{n^2} \to 0$$

almost surely, by the above working. □